

Linear Algebra — Question Bank

Chapter 2: Linear System of Equations

Topic-wise, with worked examples and fully worked-out practice solutions

How to use this question bank

This file is organised into six topics, following the same order as the source chapter. Each topic opens with a short

Quick Reference

box (definitions and results, no derivations), followed by every

Worked Example

from the source notes for that topic – shown fully solved, exactly as derived there – and then a set of

Practice Questions

. Unlike the Chapter 3 question bank, every Question here carries its own Solution directly underneath it – there is no separate solutions file. Every solution is written out step by step: no algebraic step is skipped, and at each step there is a short remark on *why* that step is legitimate or *where* an assumption (a sign choice, a free variable, a particular elementary operation) is coming from. Read the Quick Reference, work through the Worked Examples with a pencil in hand (re-deriving each line yourself before reading on), and only then attempt each Question – covering the printed Solution with a sheet of paper until you have your own answer to compare it against.

Contents

Topic 1	Linear Systems, Elementary Operations, and Gauss Elimination	2
Topic 2	Row Reduced Echelon Form and Elementary Matrices	8
Topic 3	Rank of a Matrix	14
Topic 4	Existence and Uniqueness of Solutions to $Ax = b$	18
Topic 5	Invertible Matrices	23
Topic 6	Determinants, Adjoint, and Cramer's Rule	28

Topic 1 Linear Systems, Elementary Operations, and Gauss Elimination

Quick Reference

- A **linear system** of m equations in n unknowns $x_1, \dots, x_n$ is a set of equations $a_{i1}x_1 + \dots + a_{in}x_n = b_i$ ($1 \leq i \leq m$), written in matrix form as $Ax = b$, where $A = [a_{ij}]$ is the $m \times n$ **coefficient matrix** and $[A \mid b]$ is the **augmented matrix**. The system is **homogeneous** if $b = 0$, **non-homogeneous** otherwise. For $Ax = 0$, the zero vector is always a solution (the **trivial solution**); any other solution is **non-trivial**.
- Two linear systems are **equivalent** if one can be obtained from the other by a finite sequence of **elementary operations**: (1) interchange two equations, (2) multiply an equation by a nonzero constant, (3) replace an equation by itself plus a constant multiple of another equation. **Equivalent systems have exactly the same solution set** (Theorem 2.3.4) – this is the single fact that justifies every row-reduction computation in this chapter.
- The same three operations applied to the rows of $[A \mid b]$ are the **elementary row operations**: R_{ij} (interchange rows i, j), $R_k(c)$ (multiply row k by $c \neq 0$), $R_{kj}(c)$ (replace row k by row k plus c times row j). Two matrices are **row-equivalent** if one is obtained from the other by finitely many elementary row operations.
- **Gauss elimination**: use elementary row operations to bring $[A \mid b]$ to upper-triangular (**forward elimination**), then solve by **back substitution**. At every stage the resulting system is equivalent to the original one, so it has exactly the same solutions.

Worked Examples

Example 2.2.4

Solve the linear system $x + 7y + 3z = 11$, $x + y + z = 3$, $4x + 10y - z = 13$.

Solution. Apply elementary operations one at a time, at each stage noting which operation is used and why the new system still has exactly the same solutions as the previous one (Theorem 2.3.4 guarantees this for every step below).

1. Interchange the 1st and 2nd equations (purely for convenience, so the equation with the simplest coefficients leads): $x + y + z = 3$, $x + 7y + 3z = 11$, $4x + 10y - z = 13$.
2. Subtract the 1st equation from the 2nd, and 4 times the 1st equation from the 3rd (this eliminates x from every equation except the first): $x + y + z = 3$, $6y + 2z = 8$, $6y - 5z = 1$.
3. Subtract the 2nd equation from the 3rd (eliminates y from the 3rd equation): $x + y + z = 3$, $6y + 2z = 8$, $-7z = -7$.
4. Divide the 2nd equation by 2: $x + y + z = 3$, $3y + z = 4$, $-7z = -7$; divide the 3rd by -7 : $z = 1$.

Now **back-substitute**: $z = 1$ into $3y + z = 4$ gives $y = 1$; then $x + y + z = 3$ gives $x = 3 - 1 - 1 = 1$. The solution set is $\{(x, y, z)^t = (1, 1, 1)^t\}$, a single point – the three planes meet at one point.

Example 2.3.10 (Gauss elimination – unique solution)

Solve $y + z = 2$, $2x + 3z = 5$, $x + y + z = 3$.

Solution. The augmented matrix is $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 1 & 1 & 1 & 3 \end{pmatrix}$.

1. R_{12} (the $(1, 1)$ entry is 0, so we must swap in a row with a nonzero first entry before we can use it as a pivot): $\begin{pmatrix} 2 & 0 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \end{pmatrix}$.

2. $R_1(1/2)$ (normalize the pivot to 1): $\begin{pmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \end{pmatrix}$.

3. $R_{31}(-1)$ (clear the rest of column 1 below the pivot): $\begin{pmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$.

4. $R_{32}(-1)$ (clear column 2 below the second pivot): $\begin{pmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -\frac{3}{2} & -\frac{3}{2} \end{pmatrix}$.

5. $R_3(-2/3)$ (normalize the third pivot): $\begin{pmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{pmatrix}$.

Reading off: $z = 1$; then $y + z = 2 \Rightarrow y = 1$; then $x + \frac{3}{2}z = \frac{5}{2} \Rightarrow x = 1$. Unique solution $(x, y, z)^t = (1, 1, 1)^t$.

Example 2.3.11 (Gauss elimination – infinitely many solutions)

Solve $x + y + z = 3$, $x + 2y + 2z = 5$, $3x + 4y + 4z = 11$.

Solution. Augmented matrix $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & 2 & 5 \\ 3 & 4 & 4 & 11 \end{pmatrix}$.

1. $R_{21}(-1)$, $R_{31}(-3)$: $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{pmatrix}$.

2. $R_{32}(-1)$: $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$.

The third row is entirely zero – this is *not* a contradiction, it simply means the third original equation carried no new information (it was a combination of the first two). Only two pivots appear (in columns for x and y), so z is a **free variable**: from row 2, $y = 2 - z$; from row 1, $x = 3 - y - z = 3 - (2 - z) - z = 1$. So the solution set is

$$(x, y, z)^t = (1, 2 - z, z)^t = (1, 2, 0)^t + z(0, -1, 1)^t, \quad z \in \mathbb{R} \text{ arbitrary,}$$

an infinite set (a line's worth of solutions, since the three planes intersect in a line rather than a point).

Example 2.3.12 (Gauss elimination – no solution)

Solve $x + y + z = 3$, $x + 2y + 2z = 5$, $3x + 4y + 4z = 12$ (identical to the previous system except the last constant is 12 instead of 11).

Solution. Augmented matrix $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & 2 & 5 \\ 3 & 4 & 4 & 12 \end{pmatrix}$; the same two operations $R_{21}(-1)$, $R_{31}(-3)$ give

$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 3 \end{pmatrix}$, and then $R_{32}(-1)$ gives $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$. The third row reads $0x + 0y + 0z = 1$,

which is false for *every* choice of x, y, z – no values can make $0 = 1$. Since this row-reduced system is equivalent to the original (Theorem 2.3.4), and it has no solutions, the original system has no solution either. (This one-off change in a single constant, from 11 to 12, is exactly what tips the system from "a line of solutions" to "no solution at all" – a good illustration of how delicately consistency can depend on the data.)

Practice Questions

Question 1.1

Solve, by Gauss elimination, $x + y + z = 6$, $2x + y - z = 1$, $x - y + 2z = 5$.

Solution. Write the augmented matrix:

$$[A | b] = \begin{pmatrix} 1 & 1 & 1 & 6 \\ 2 & 1 & -1 & 1 \\ 1 & -1 & 2 & 5 \end{pmatrix}.$$

Step 1 (clear column 1 below the first pivot). The $(1,1)$ entry is already 1, so it can serve as the pivot without any swap or scaling. Apply $R_{21}(-2)$ (subtract $2 \times$ row 1 from row 2, to kill the leading 2) and $R_{31}(-1)$ (subtract row 1 from row 3, to kill the leading 1):

$$R_2 \rightarrow R_2 - 2R_1 = (2-2, 1-2, -1-2, 1-12) = (0, -1, -3, -11), \quad R_3 \rightarrow R_3 - R_1 = (0, -2, 1, -1).$$

$$\begin{pmatrix} 1 & 1 & 1 & 6 \\ 0 & -1 & -3 & -11 \\ 0 & -2 & 1 & -1 \end{pmatrix}.$$

Why this is legitimate: replacing a row by itself plus a multiple of another row is elementary operation type 3, and Theorem 2.3.4 guarantees the new system has exactly the same solution set as the old one – so we may keep working with this simpler matrix instead of the original.

Step 2 (clear column 2 below the second pivot). The $(2,2)$ entry is $-1 \neq 0$, so it can be a pivot. Apply $R_{32}(-2)$, i.e. replace row 3 by row 3 minus $2 \times$ (row 2's ratio) – concretely, since we want to cancel the -2 in row 3 using the -1 in row 2, the correct multiplier is c with $-2 + c(-1) = 0 \Rightarrow c = -2$, i.e. $R_3 \rightarrow R_3 - 2R_2$:

$$R_3 \rightarrow R_3 - 2R_2 = (0 - 0, -2 - 2(-1), 1 - 2(-3), -1 - 2(-11)) = (0, 0, 7, 21).$$

$$\begin{pmatrix} 1 & 1 & 1 & 6 \\ 0 & -1 & -3 & -11 \\ 0 & 0 & 7 & 21 \end{pmatrix}.$$

This is already upper triangular, so forward elimination is complete.

Step 3 (back substitution). The third row reads $7z = 21$, so $z = 3$. *Why we can just read this off:* a single nonzero entry in the last row, in the z -column, means this equation involves only z – there is nothing left to eliminate.

Substitute $z = 3$ into the second row's equation $-y - 3z = -11$: $-y - 9 = -11 \Rightarrow -y = -2 \Rightarrow y = 2$.

Substitute $y = 2, z = 3$ into the first row's equation $x + y + z = 6$: $x + 2 + 3 = 6 \Rightarrow x = 1$.

Solution: $(x, y, z)^t = (1, 2, 3)^t$, and since forward elimination produced a pivot in every column (three pivots for three unknowns, no free variable remained), this solution is unique.

Check (always verify against the *original* equations, not the row-reduced ones, since that is the real test of correctness): $1 + 2 + 3 = 6\checkmark$; $2(1) + 2 - 3 = 1\checkmark$; $1 - 2 + 2(3) = 1 - 2 + 6 = 5\checkmark$.

Question 1.2

Solve, by Gauss elimination, $x + y + z = 6$, $2x + y + 3z = 10$, $3x + 2y + 4z = 16$.

Solution. Augmented matrix:

$$\begin{pmatrix} 1 & 1 & 1 & 6 \\ 2 & 1 & 3 & 10 \\ 3 & 2 & 4 & 16 \end{pmatrix}.$$

Step 1. $R_{21}(-2)$: $(2 - 2, 1 - 2, 3 - 2, 10 - 12) = (0, -1, 1, -2)$. $R_{31}(-3)$: $(3 - 3, 2 - 3, 4 - 3, 16 - 18) = (0, -1, 1, -2)$.

$$\begin{pmatrix} 1 & 1 & 1 & 6 \\ 0 & -1 & 1 & -2 \\ 0 & -1 & 1 & -2 \end{pmatrix}.$$

Step 2. Row 3 is now *identical* to row 2, so $R_{32}(-1)$ (subtract row 2 from row 3) gives an all-zero row:

$$\begin{pmatrix} 1 & 1 & 1 & 6 \\ 0 & -1 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Why a zero row appears here, and what it means: the third original equation ($3x + 2y + 4z = 16$) turned out to equal the sum of the first equation ($x + y + z = 6$) and the second ($2x + y + 3z = 10$): $6 + 10 = 16\checkmark$ and coefficient-wise $(1, 1, 1) + (2, 1, 3) = (3, 2, 4)\checkmark$. So the third equation carried *no independent information* – it was already a consequence of the first two – and row reduction has exposed exactly this redundancy as a $0 = 0$ row, which is always true and imposes no constraint.

Only two pivots (columns 1 and 2) were produced for three unknowns, so one variable is free; take z to be that free variable (it is the one whose column has no pivot). From row 2: $-y + z = -2 \Rightarrow y = z + 2$. From row 1: $x + y + z = 6 \Rightarrow x = 6 - y - z = 6 - (z + 2) - z = 4 - 2z$.

Solution set:

$$(x, y, z)^t = (4 - 2z, z + 2, z)^t = (4, 2, 0)^t + z(-2, 1, 1)^t, \quad z \in \mathbb{R} \text{ arbitrary.}$$

This is an infinite family (a line in $\mathbb{R}^3$), because the system, despite having three equations, secretly only carries two independent pieces of information.

Check the general solution satisfies all three original equations identically in z (this checks the whole family at once, not just one point): first: $(4 - 2z) + (z + 2) + z = 6\checkmark$ (the z 's

cancel); second: $2(4-2z)+(z+2)+3z = 8-4z+z+2+3z = 10\checkmark$; third: $3(4-2z)+2(z+2)+4z = 12-6z+2z+4+4z = 16\checkmark$.

Question 1.3

Show that $x + y + z = 6$, $2x + y + 3z = 10$, $3x + 2y + 4z = 17$ has *no* solution. (Notice this is the same system as Question 1.2, except the last constant is 17 instead of 16 – compare with how Examples 2.3.11/2.3.12 differed by a single constant.)

Solution. Augmented matrix:

$$\begin{pmatrix} 1 & 1 & 1 & 6 \\ 2 & 1 & 3 & 10 \\ 3 & 2 & 4 & 17 \end{pmatrix}.$$

Exactly the same row operations as Question 1.2 apply to columns 1–2 (they only depend on the coefficient matrix, which hasn't changed, not on the last column):

$R_{21}(-2)$: row 2 becomes $(0, -1, 1, 10 - 12) = (0, -1, 1, -2)$ (identical to before, since the constant 10 didn't change).

$R_{31}(-3)$: row 3 becomes $(0, -1, 1, 17 - 18) = (0, -1, 1, -1)$ (*this* constant did change: $17 - 3(6) = -1$ instead of -2).

$$\begin{pmatrix} 1 & 1 & 1 & 6 \\ 0 & -1 & 1 & -2 \\ 0 & -1 & 1 & -1 \end{pmatrix}.$$

Now $R_{32}(-1)$ (subtract row 2 from row 3): the coefficient part of row 3, $(-1, 1) - (-1, 1) = (0, 0)$, still cancels exactly as before (it only depended on the coefficients, unchanged from Question 1.2) – *but* the constant is now $-1 - (-2) = 1$, not 0:

$$\begin{pmatrix} 1 & 1 & 1 & 6 \\ 0 & -1 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The last row reads $0x + 0y + 0z = 1$, i.e. $0 = 1$, which is false regardless of x, y, z . Since this system is equivalent to the original (each step used only elementary row operations, which by Theorem 2.3.4 preserve the solution set exactly), and the reduced system plainly has no solution, **the original system has no solution.**

Why changing one constant flips "infinitely many" into "none": in Question 1.2 the third equation happened to equal exactly (1st equation) + (2nd equation), giving $6 + 10 = 16$ consistently on the right-hand side too. Here the coefficients on the left are unchanged (so the same dependency $3x + 2y + 4z = (x + y + z) + (2x + y + 3z)$ still holds automatically for the left-hand sides), but the prescribed right-hand side $17 \neq 6 + 10 = 16$ – so the third equation demands something the first two, added together, flatly contradict. This is exactly Theorem 2.6.1's trichotomy: the coefficient matrix has rank 2 in both Question 1.2 and here, but the augmented matrix's rank jumps to 3 here (because the extra row is only a coefficient-dependency, not a constant-dependency), and $\text{rank}(A) < \text{rank}([A | b])$ is precisely the no-solution case.

Question 1.4

One of the three elementary row operations is “multiply row k by a nonzero constant c ”, denoted $R_k(c)$. Prove that if $Cx = d$ is obtained from $Ax = b$ by this single operation, then $Ax = b$ and $Cx = d$ have exactly the same set of solutions. (The source notes prove this for operation type 3, replacing a row by itself plus a multiple of another row; this question asks for the analogous proof for type 2, scaling a single row.)

Solution. Say the operation scales the k th equation: originally $a_{k1}x_1 + \cdots + a_{kn}x_n = b_k$, and after the operation this becomes $ca_{k1}x_1 + \cdots + ca_{kn}x_n = cb_k$ (every coefficient, and the constant, multiplied by c); all other equations are left untouched, so $Cx = d$ agrees with $Ax = b$ in every row except the k th.

Direction 1: every solution of $Ax = b$ solves $Cx = d$. Let $(\alpha_1, \dots, \alpha_n)$ satisfy $Ax = b$. All equations except the k th are literally unchanged between the two systems, so those are automatically satisfied by $(\alpha_1, \dots, \alpha_n)$ in $Cx = d$ too – there is nothing to check there. For the k th equation, we know $a_{k1}\alpha_1 + \cdots + a_{kn}\alpha_n = b_k$ holds (since (α_i) solves $Ax = b$). *Why we may multiply both sides of a true equation by the same constant and keep it true:* this is just a property of real numbers – if $p = q$ then $cp = cq$ for any c , since multiplication by a fixed number is a well-defined operation on both sides of an equality. Applying this with c (the same nonzero scalar used in the row operation):

$$c(a_{k1}\alpha_1 + \cdots + a_{kn}\alpha_n) = cb_k \implies (ca_{k1})\alpha_1 + \cdots + (ca_{kn})\alpha_n = cb_k,$$

which is exactly the k th equation of $Cx = d$. So $(\alpha_1, \dots, \alpha_n)$ satisfies every equation of $Cx = d$.

Direction 2: every solution of $Cx = d$ solves $Ax = b$. Let $(\beta_1, \dots, \beta_n)$ satisfy $Cx = d$. Again all equations except the k th are shared between the systems, so those hold for $Ax = b$ automatically. For the k th equation, we know $(ca_{k1})\beta_1 + \cdots + (ca_{kn})\beta_n = cb_k$. *This is exactly where the hypothesis “ $c \neq 0$ ” is used, and it is the only place it is needed:* since $c \neq 0$, we are allowed to divide both sides by c (multiply by $1/c$, which exists precisely because $c \neq 0$):

$$\frac{1}{c}[(ca_{k1})\beta_1 + \cdots + (ca_{kn})\beta_n] = \frac{1}{c}(cb_k) \implies a_{k1}\beta_1 + \cdots + a_{kn}\beta_n = b_k,$$

which is exactly the k th equation of $Ax = b$. So $(\beta_1, \dots, \beta_n)$ satisfies every equation of $Ax = b$. Since each solution set contains the other, they are equal: $Ax = b$ and $Cx = d$ have exactly the same solutions. ■

Remark (why $c \neq 0$ is essential, not a technicality). If c were allowed to be 0, the “operation” would replace the k th equation by $0 = 0$, which is satisfied by *every* $(x_1, \dots, x_n)$ – so $Cx = d$ could gain extra solutions that $Ax = b$ never had (Direction 2 above literally breaks down at the division-by- c step). This is exactly why the definition of the operation insists on a *nonzero* constant.

Topic 2 Row Reduced Echelon Form and Elementary Matrices

Quick Reference

- A matrix C is in **row reduced form** if (1) the first nonzero entry in each row is 1, and (2) the column containing that 1 has all other entries equal to 0. It is in **row reduced echelon form (RREF)** if, in addition, (3) all-zero rows come below all nonzero rows, and (4) the leading 1's move strictly left-to-right as you go down the rows.
- The first nonzero entry in a row of a row-reduced matrix is its **leading term**; the column it sits in is a **leading** (or **pivot**) **column**. If a system's augmented matrix $[A \mid b]$ row-reduces to $[C \mid d]$ with r pivot columns among the first n , the corresponding variables are **basic**; the other $n - r$ variables are **free**.
- **Theorem 2.4.11**: the RREF of a matrix is unique – no matter which sequence of row operations you use to get there, you always land on the same final matrix.
- An **elementary matrix** is obtained by applying exactly one elementary row operation to I_n : E_{ij} (swap rows i, j of I_n), $E_k(c)$ (scale row k of I_n by $c \neq 0$), $E_{ij}(c)$ (add c times row j of I_n to row i). **Key fact** (Exercise 2.4.18.1): for any matrix A , applying an elementary row operation e to A gives exactly EA , where $E = e(I)$ – i.e., every row operation is secretly a left-multiplication by the corresponding elementary matrix.
- Right-multiplying A by an elementary matrix performs the *column* version of the same operation on A (Example 2.4.17): $A E_{ij}$ swaps columns i, j of A , etc.
- Two matrices A, B are row-equivalent iff $B = PA$ for some invertible P that is a product of elementary matrices.

Worked Examples

Example 2.4.7

Let $A = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$, both already in row reduced form (check:

the first nonzero entry of each nonzero row is 1, and each such column is otherwise all zero). Find their row reduced echelon forms.

Solution. A already satisfies condition (2) of RREF (zero rows below nonzero rows) and condition (3) trivially since it has only two nonzero rows and their leading columns (2, then 3) already increase left to right. So A itself is already in RREF.

B 's nonzero rows are not in the right order: row 1's leading term is in column 3, but row 2's leading term is in column 1 – column $1 < 3$, so the leading columns do *not* increase top-to-bottom as required. Simply reordering the rows (an interchange, an elementary row operation, so the row space and solution behaviour are unaffected) by moving the row with the smallest

leading column to the top fixes this:

$$\text{RREF}(B) = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

This illustrates that “row reduced” and “row reduced *echelon*” are different conditions: every RREF matrix is row-reduced, but (as B shows) not every row-reduced matrix is already in echelon order.

Example 2.4.15 (row operations as left multiplication)

Let $A = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 2 & 0 & 3 & 4 \\ 3 & 4 & 5 & 6 \end{pmatrix}$. Show that interchanging rows 2 and 3 of A is the same as left-multiplying A by E_{23} .

Solution. Applying R_{23} directly to A gives $\begin{pmatrix} 1 & 2 & 3 & 0 \\ 3 & 4 & 5 & 6 \\ 2 & 0 & 3 & 4 \end{pmatrix}$.

Now build $E_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ (apply R_{23} to I_3 itself) and multiply:

$$E_{23}A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 0 \\ 2 & 0 & 3 & 4 \\ 3 & 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 3 & 4 & 5 & 6 \\ 2 & 0 & 3 & 4 \end{pmatrix}.$$

(Row i of $E_{23}A$ is (row i of E_{23}) $\cdot$ columns of A , i.e. a combination of A 's rows with the weights in row i of E_{23} – since row 2 of E_{23} is $(0, 0, 1)$, row 2 of $E_{23}A$ is exactly row 3 of A , and symmetrically for row 3.) The two results agree exactly, confirming $R_{23}(A) = E_{23}A$.

Example 2.4.17 (column operations as right multiplication)

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \\ 3 & 4 & 5 \end{pmatrix}$ and let f be the elementary column operation that interchanges columns 2 and 3 of A . Show $f(A) = AE_{23}$.

Solution. Applying f directly: swap columns 2, 3 of A to get $f(A) = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 3 & 0 \\ 3 & 5 & 4 \end{pmatrix}$.

Now compute AE_{23} with $E_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$: column j of AE_{23} is $A \cdot$ (column j of E_{23}), i.e. a combination of A 's *columns* with the weights in column j of E_{23} . Column 2 of E_{23} is $(0, 0, 1)^t$, so column 2 of AE_{23} is exactly column 3 of A ; column 3 of E_{23} is $(0, 1, 0)^t$, so column 3 of AE_{23}

is column 2 of A . This gives $AE_{23} = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 3 & 0 \\ 3 & 5 & 4 \end{pmatrix}$, matching $f(A)$ exactly. So multiplying on the *right* by an elementary matrix performs the corresponding operation on *columns*, mirroring how left multiplication performs it on rows.

Practice Questions

Question 2.1

Find the row reduced echelon form of $M = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 3 & 1 & 0 \\ 1 & 0 & 7 & 5 \end{pmatrix}$, and identify the pivot columns together with the basic and free variables (treating M as the coefficient part of a system in 4 unknowns).

Solution.

Step 1 (use row 1 to clear column 1 below the pivot). The $(1,1)$ entry is already 1, so no swap or scaling is needed before using it as a pivot. Apply $R_{21}(-2)$ and $R_{31}(-1)$:

$$R_2 \rightarrow R_2 - 2R_1 = (2 - 2, 3 - 2, 1 - 4, 0 - 2) = (0, 1, -3, -2), \quad R_3 \rightarrow R_3 - R_1 = (0, -1, 5, 4).$$

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -2 \\ 0 & -1 & 5 & 4 \end{pmatrix}.$$

Why this is the correct multiplier: to eliminate the 2 in position $(2,1)$ using the pivot 1 in position $(1,1)$, we need $2 + c(1) = 0 \Rightarrow c = -2$; similarly the 1 in position $(3,1)$ needs $1 + c(1) = 0 \Rightarrow c = -1$.

Step 2 (use row 2 to clear column 2 below the pivot). The $(2,2)$ entry is now 1, a usable pivot. To cancel the -1 in position $(3,2)$: $-1 + c(1) = 0 \Rightarrow c = 1$, i.e. $R_{32}(1)$ (add row 2 to row 3):

$$R_3 \rightarrow R_3 + R_2 = (0 - 0, -1 + 1, 5 - 3, 4 - 2) = (0, 0, 2, 2).$$

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 2 & 2 \end{pmatrix}.$$

Step 3 (normalize the third pivot). $R_3(1/2)$: row 3 becomes $(0, 0, 1, 1)$.

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Forward elimination is done (upper triangular with three 1's on the pivots). Now do **back substitution** – work from the bottom pivot upward, using each pivot row to clear *above* itself, since RREF additionally requires zeros *above* each leading 1, not just below.

Step 4 (use row 3 to clear column 3 above the pivot). Column 3 has entries 2 (row 1) and -3 (row 2) above the pivot 1. Cancel with $R_{13}(-2)$ and $R_{23}(3)$:

$$R_1 \rightarrow R_1 - 2R_3 = (1 - 0, 1 - 0, 2 - 2, 1 - 2) = (1, 1, 0, -1),$$

$$R_2 \rightarrow R_2 + 3R_3 = (0, 1, 0, -2 + 3) = (0, 1, 0, 1).$$

$$\begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Step 5 (use row 2 to clear column 2 above the pivot). Column 2 has a 1 in row 1 above the pivot. $R_{12}(-1)$:

$$R_1 \rightarrow R_1 - R_2 = (1 - 0, 1 - 1, 0 - 0, -1 - 1) = (1, 0, 0, -2).$$

$$\text{RREF}(M) = \begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

This is now genuinely in RREF: every leading 1 has zeros both above and below it in its column, the leading columns (1, 2, 3) strictly increase left to right, and there is no all-zero row to worry about ordering.

Reading off the structure. The leading terms sit in columns 1, 2, 3 – these are the **pivot columns**, so (treating the four columns as belonging to variables x_1, x_2, x_3, x_4) x_1, x_2, x_3 are the **basic variables** and x_4 (the one column with no pivot) is the **free variable**. *Why exactly one variable is free here:* there are 3 nonzero rows (i.e. rank 3) and 4 columns, and Remark 2.4.5 guarantees $n - r = 4 - 3 = 1$ free variable – consistent with what we found by direct inspection.

Question 2.2

Let e be the elementary row operation “ $R_{31}(2)$ ” (replace row 3 by row 3 plus $2 \times$ row 1), and let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Write down $E = e(I_3)$ explicitly, compute EA , compute $e(A)$ directly, and confirm $e(A) = EA$ (a specific instance of Exercise 2.4.18.1’s general claim).

Solution.

Building E . By definition $E = e(I_3)$, so apply $R_{31}(2)$ to $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$: row 3 becomes

(row 3) + $2 \times$ (row 1) = $(0 + 2(1), 0 + 2(0), 1 + 2(0)) = (2, 0, 1)$, and rows 1, 2 are untouched (the operation only names row 3 as the one being replaced):

$$E = E_{31}(2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}.$$

Computing EA . Row i of EA is (row i of E) times the columns of A – i.e. the combination of A ’s rows with the weights sitting in row i of E .

- Row 1 of E is $(1, 0, 0)$, so row 1 of EA is exactly row 1 of A : $(1, 0, 2)$.
- Row 2 of E is $(0, 1, 0)$, so row 2 of EA is exactly row 2 of A : $(0, 1, 1)$.
- Row 3 of E is $(2, 0, 1)$, so row 3 of EA is $2 \times$ (row 1 of A) + $0 \times$ (row 2 of A) + $1 \times$ (row 3 of A) = $2(1, 0, 2) + (1, 1, 0) = (2, 0, 4) + (1, 1, 0) = (3, 1, 4)$.

$$EA = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 3 & 1 & 4 \end{pmatrix}.$$

Computing $e(A)$ directly. Apply $R_{31}(2)$ to A itself: row 3 becomes (row 3 of A) $+2 \times$ (row 1 of A) $= (1, 1, 0) + 2(1, 0, 2) = (1, 1, 0) + (2, 0, 4) = (3, 1, 4)$; rows 1, 2 unchanged.

$$e(A) = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 3 & 1 & 4 \end{pmatrix}.$$

Conclusion. $e(A) = EA$ exactly, confirming the general fact in this instance. *Why this always works, not just here:* row 3 of EA is, by the mechanics of matrix multiplication, the linear combination of A 's rows with coefficients read off from row 3 of E – and by construction $E = e(I)$ has *exactly* those coefficients in row 3 (since applying e to I records, in each row of E , precisely which combination of the standard basis rows produces that row of $e(I)$). Multiplying that same combination against A 's actual rows (instead of I 's rows) reproduces the identical operation on A 's rows – so $e(A) = EA$ holds for *any* matrix A with the same number of rows as E , not just this particular one.

Question 2.3

Prove: two $m \times n$ matrices A, B are row-equivalent if and only if $B = PA$ for some invertible matrix P that is a product of elementary matrices (Exercise 2.4.18.3).

Solution.

($\Rightarrow$) Suppose A, B are row-equivalent. By definition this means B is obtained from A by a finite sequence of elementary row operations, say $e_1, e_2, \dots, e_k$ applied in that order: $e_1(A)$ first, then e_2 applied to that result, and so on, with $B = e_k(e_{k-1}(\dots e_1(A) \dots))$.

By the key fact in the Quick Reference (Exercise 2.4.18.1, verified concretely in Question 2.2), applying e_1 to A is the same as left-multiplying by $E_1 := e_1(I_m)$: $e_1(A) = E_1 A$. Applying e_2 to *that* matrix is left-multiplication by $E_2 := e_2(I_m)$: $e_2(E_1 A) = E_2 E_1 A$. Continuing this way (each subsequent operation left-multiplies the *running product so far* by its own elementary matrix, since the fact “ $e(M) = e(I)M$ ” applies no matter what matrix M currently is), after all k steps:

$$B = E_k E_{k-1} \cdots E_2 E_1 A.$$

Set $P := E_k E_{k-1} \cdots E_2 E_1$, a product of elementary matrices, so $B = PA$. *Why P is automatically invertible:* every elementary matrix is invertible (swapping rows twice undoes itself, so $E_{ij}^{-1} = E_{ij}$; scaling by c is undone by scaling by $1/c$, so $E_k(c)^{-1} = E_k(1/c)$; adding c times row j to row i is undone by subtracting it back, so $E_{ij}(c)^{-1} = E_{ij}(-c)$ – these are exactly the “inverse operations” identified in Exercise 2.3.6), and a product of invertible matrices is invertible (with inverse $E_1^{-1} E_2^{-1} \cdots E_k^{-1}$, in reverse order). So P is invertible.

($\Leftarrow$) Suppose $B = PA$ with $P = F_k F_{k-1} \cdots F_1$ a product of elementary matrices $F_1, \dots, F_k$. We must show B is obtained from A by a finite sequence of elementary row operations. Reading the product from the *inside out*: $F_1 A$ is, by the key fact again, exactly $f_1(A)$, where f_1 is the elementary row operation corresponding to F_1 (every elementary matrix, being $e(I)$ for some operation e , is itself the “code” for that operation). Then $F_2(F_1 A) = F_2 f_1(A) = f_2(f_1(A))$, applying the row operation f_2 corresponding to F_2 to the

previous result. Continuing up the chain,

$$B = PA = F_k F_{k-1} \cdots F_1 A = f_k(f_{k-1}(\cdots f_1(A) \cdots)),$$

i.e. B is exactly the result of applying the sequence of elementary row operations $f_1, f_2, \dots, f_k$ (in that order) to A . By definition, this makes A, B row-equivalent.

Since each direction holds, A, B are row-equivalent $\iff B = PA$ for some invertible product-of-elementary-matrices P . ■

Is P unique? No. For instance $P = I$ always works when $B = A$ (trivially row-equivalent to itself via zero operations), but if $A = B$ has any repeated row, other nontrivial products of elementary matrices can also fix A (e.g. an operation that changes a row and then its exact inverse operation, composed together, give $P = I$ again but written as a nontrivial product $E^{-1}E$). More substantially, if A has fewer than m nonzero rows in its RREF, there is “leftover room” in the row operations used on the zero rows, so different sequences of eliminations can produce different valid P ’s with the same effect on A . Uniqueness of P would require A itself to be invertible (so that $P = BA^{-1}$ is pinned down completely); in general it is not.

Question 2.4

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$. Find the elementary matrix $E_2(-3)$ of the appropriate size for right multiplication, compute $AE_2(-3)$, and describe in words which column operation this performs – then verify directly by performing that operation on A by hand.

Solution. Since A has 2 columns, right-multiplication needs a 2×2 elementary matrix. By definition, $E_2(-3) = e(I_2)$ where e is “multiply row 2 by -3 ”: apply this to $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, changing only row 2 $(0, 1) \rightarrow (0, -3)$:

$$E_2(-3) = \begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}.$$

Computing $AE_2(-3)$. Column j of $AE_2(-3)$ is A times column j of $E_2(-3)$ – i.e. a combination of A ’s *columns*, weighted by the entries in column j of $E_2(-3)$. Column 1 of $E_2(-3)$ is $(1, 0)^t$, so column 1 of $AE_2(-3)$ is $1 \times (\text{col 1 of } A) + 0 \times (\text{col 2 of } A) = \text{column 1 of } A$, unchanged. Column 2 of $E_2(-3)$ is $(0, -3)^t$, so column 2 of $AE_2(-3)$ is $0 \times (\text{col 1}) + (-3) \times (\text{col 2 of } A) = -3 \times \text{column 2 of } A$:

$$AE_2(-3) = \begin{pmatrix} 1 & 2 \cdot (-3) \\ 3 & 4 \cdot (-3) \\ 5 & 6 \cdot (-3) \end{pmatrix} = \begin{pmatrix} 1 & -6 \\ 3 & -12 \\ 5 & -18 \end{pmatrix}.$$

In words: right-multiplying by $E_2(-3)$ performs the elementary *column* operation “multiply column 2 by -3 ” on A – this mirrors exactly how left-multiplying by $E_k(c)$ scales *row* k , with the operation now acting on the second index (columns) instead of the first (rows) because right multiplication combines A ’s columns rather than its rows.

Direct verification. Multiplying column 2 of $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$ by -3 by hand: column 2 is

$(2, 4, 6)^t$, so $-3 \times (2, 4, 6)^t = (-6, -12, -18)^t$, giving $\begin{pmatrix} 1 & -6 \\ 3 & -12 \\ 5 & -18 \end{pmatrix}$ – identical to $AE_2(-3)$ computed above. ✓

Topic 3 Rank of a Matrix

Quick Reference

- The **row-rank** of a matrix is the number of nonzero rows in its row reduced form; row-equivalent matrices automatically have the same row-rank (since row operations are reversible, so a row-reduced form reachable from A is reachable from B too, and RREF is unique).
- Analogously, a **column-rank** can be defined using elementary column operations. It is a theorem (proved below, Question 3.1) that row-rank always equals column-rank; this common value is simply called the **rank** of A , written $\text{rank}(A)$.
- **Theorem 2.5.7**: if $\text{rank}(A) = r$, there exist invertible P (a product of elementary row-matrices) and Q (a product of elementary column-matrices) with $PAQ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$.
- **Corollary 2.6.3** (stated here for reference, proved under Topic 4): $Ax = 0$ has a non-trivial solution iff $\text{rank}(A) < n$ (the number of columns/unknowns).

Worked Examples

Example 2.5.3

Determine the row-rank of (1) $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ and (2) $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 0 \end{pmatrix}$.

Solution. (1) $R_{21}(-2), R_{31}(-1)$: $\begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & -1 & 1 \end{pmatrix}$. Then $R_2(-1), R_{32}(1)$: $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$. Then

$R_3(1/2), R_{12}(-2)$: $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. Then $R_{23}(-1), R_{13}(1)$: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3$. Three nonzero rows, so $\text{row-rank}(A) = 3$.

(2) $R_{21}(-2), R_{31}(-1)$: $\begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix}$. Then $R_2(-1), R_{32}(1)$ (subtract row 2 from row 3, since

they're now identical): $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. Only two nonzero rows: $\text{row-rank}(A) = 2$ – the change

from 2 (last entry) to 0 in the last row of the original matrix was enough to make the third row a combination of the first two.

Practice Questions

Question 3.1

Prove that $\text{rank}(A) = \text{rank}(A^t)$ for every matrix A (Exercise 2.5.8.2).

Solution. Write $\text{rank}(A) = r$ for the row-rank of A (dimension of its row space, viewed as a subspace of $\mathbb{R}^n$ if A is $m \times n$), and let c denote the **column-rank** of A , the dimension of its column space (a subspace of $\mathbb{R}^m$). We show $r \leq c$ and $c \leq r$, which forces $r = c$; then note $\text{rank}(A^t)$ equals the row-rank of A^t , which is exactly the column-rank of A (transposing turns columns into rows) – so proving row-rank=column-rank for every matrix is exactly what is needed.

Step 1: $r \leq c$. Let $b_1, \dots, b_c \in \mathbb{R}^m$ be a basis of the column space of A (there are c of them since c is that space's dimension). Every column a_j of A lies in the column space, so it can be written as a combination of this basis: $a_j = \sum_{k=1}^c m_{kj} b_k$ for some scalars m_{kj} . *Why this representation is possible for every single column:* by definition, the column space is the span of A 's own columns, and $b_1, \dots, b_c$ span that same space (they're a basis of it), so every column of A – being an element of that span – is automatically some combination of $b_1, \dots, b_c$.

Collect the b_k 's as the columns of an $m \times c$ matrix $B = [b_1 \ \cdots \ b_c]$, and collect the coefficients m_{kj} into a $c \times n$ matrix $M = [m_{kj}]$. The equation $a_j = \sum_k m_{kj} b_k$ says exactly that column j of A equals column j of BM (by the definition of matrix multiplication, column j of BM is B times column j of M , i.e. $\sum_k m_{kj} b_k$). Since this holds for every column $j = 1, \dots, n$, we get the factorization

$$A = BM.$$

Now look at the *rows* of A : row i of $A = BM$ is (row i of B) times M , i.e. $\sum_{k=1}^c B_{ik} (\text{row } k \text{ of } M)$ – a linear combination of the c rows of M . *Why this is exactly what row-by-matrix multiplication means:* the i th row of a product BM is obtained by combining the rows of M with weights taken from the i th row of B , which is a standard (and easily checked entrywise) fact about matrix multiplication. So **every one of A 's m rows is a linear combination of the same c rows of M** . This means the row space of A is contained in the row space of M , and since M has only c rows, its row space has dimension at most c . Therefore

$$r = \dim(\text{row space of } A) \leq \dim(\text{row space of } M) \leq c.$$

Step 2: $c \leq r$. Apply the exact same argument (Step 1) to A^t instead of A . The row-rank of A^t is, by definition, the column-rank of A , i.e. c ; and the column-rank of A^t is the row-rank of A , i.e. r . Step 1 applied to A^t says: (row-rank of A^t) $\leq$ (column-rank of A^t), i.e. $c \leq r$.

Combining. From Step 1, $r \leq c$; from Step 2, $c \leq r$. Two numbers each $\leq$ the other must be equal: $r = c$. That is, $\text{row-rank}(A) = \text{column-rank}(A)$ for every matrix A – and applying this equality to A^t in place of A gives $\text{rank}(A^t) = \text{row-rank}(A^t) = \text{column-rank}(A^t) = \text{row-rank}(A) = \text{rank}(A)$. ■

Why this two-sided “sandwich” trick is needed (rather than proving $r = c$ directly): the factorization $A = BM$ in Step 1 only directly bounds *row-rank* by *column-rank* (in one direction). Running the identical argument on the transpose swaps the roles of rows and columns, handing us the reverse inequality for free – without having to invent a second, different argument.

Question 3.2

Let A be an $n \times n$ matrix with $\text{rank}(A) = n$. Prove that A is row-equivalent to I_n (Exercise 2.5.8.3).

Solution. Let $C = \text{RREF}(A)$, the row reduced echelon form of A (which exists and is unique by Theorem 2.4.11, and is row-equivalent to A by construction, since it is reached from A purely by elementary row operations).

By definition, $\text{rank}(A)$ is the number of nonzero rows of C ; we are given this number is n . But C is an $n \times n$ matrix (row reduction never changes the shape of a matrix), and it has only n rows in total – so **every one of C 's n rows is nonzero**. Consequently C has n leading terms, one per row, sitting in n leading columns $i_1 < i_2 < \dots < i_n$ (strictly increasing, by the defining property of RREF that leading columns move strictly left to right as you go down the rows).

Why $i_k = k$ for every k is forced. There are exactly n columns available in total (columns 1 through n), and we need to choose n of them, $i_1 < i_2 < \dots < i_n$, all strictly increasing. But choosing n strictly increasing values out of only n available slots leaves no room to skip any: if even one i_k were $> k$ (skipped ahead), there would not be enough remaining columns after it to fit the remaining $n - k$ still-increasing indices $i_{k+1} < \dots < i_n$ into the columns from $i_k + 1$ up to n (only $n - i_k < n - k$ columns would remain, too few for $n - k$ further strictly increasing choices). The only way to fit n strictly increasing integers into $\{1, \dots, n\}$ is the unique increasing list itself: $i_1 = 1, i_2 = 2, \dots, i_n = n$.

So the leading term of row k sits exactly in column k , for every $k = 1, \dots, n$. Since C is in row reduced form, the leading term itself is 1, and every *other* entry in that same column is 0 – in particular, every entry in column k outside row k is 0. Reading this off for every column $k = 1, \dots, n$ simultaneously: $C_{kk} = 1$ for all k , and $C_{ij} = 0$ whenever $i \neq j$ (since the entry in row i , column $j = i_j = j$, with $i \neq j$, is exactly one of the "other entries" forced to zero by the row-reduced condition applied to column j). This is precisely the identity matrix: $C = I_n$.

Since $C = \text{RREF}(A) = I_n$ is obtained from A purely by elementary row operations, A is row-equivalent to I_n . ■

Question 3.3

Find the rank of $M = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 4 \\ -1 & -2 & 1 & 1 \\ 3 & 6 & 1 & 5 \end{pmatrix}$.

Solution.

Step 1. Column 1 has a nonzero entry in row 1 already, so use it as the pivot without swapping. Clear below it: $R_{21}(-2)$: row 2 $\rightarrow (2 - 2, 4 - 4, 1 - 0, 4 - 2) = (0, 0, 1, 2)$. $R_{31}(1)$ (since we need $-1 + c(1) = 0 \Rightarrow c = 1$): row 3 $\rightarrow (-1 + 1, -2 + 2, 1 + 0, 1 + 1) = (0, 0, 1, 2)$. $R_{41}(-3)$: row 4 $\rightarrow (3 - 3, 6 - 6, 1 - 0, 5 - 3) = (0, 0, 1, 2)$.

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix}.$$

Note already: rows 2, 3, 4 have become identical after this single step – a strong hint that the rank will turn out to be small.

Step 2. Column 2 is entirely zero below row 1 (the $(1, 2)$ entry 2 is nonzero, but there is no other nonzero entry in column 2 to use as a second pivot – rows 2, 3, 4 all have a 0 in column 2), so column 2 contributes *no new pivot*; move on to column 3 for the second pivot. Row 2 already has a nonzero entry (1) in column 3, so use it as pivot 2 directly (no swap needed). Clear the rest of column 3: $R_{32}(-1)$ (subtract row 2 from row 3, since they're identical, giving an all-zero row): row 3 $\rightarrow (0, 0, 0, 0)$. $R_{42}(-1)$: row 4 $\rightarrow (0, 0, 0, 0)$ (identical reasoning, since row 4 was also identical to row 2).

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

This is already in row reduced echelon form (leading 1's in columns 1, 3, each column otherwise clear, zero rows at the bottom, leading columns 1 < 3 increasing).

Conclusion. There are exactly 2 nonzero rows, so $\text{rank}(M) = 2$. *Why the answer is so much smaller than the 4 rows we started with:* rows 2, 3, 4 of the original matrix were secretly all multiples/combinations of the same underlying vector once row 1 was subtracted out appropriately (each equalled $(0, 0, 1, 2)$ after clearing column 1), so only one genuinely new direction was contributed beyond row 1 itself.

Question 3.4

If A and B are matrices of the same size (so that $A + B$ is defined), prove that $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.

Solution. Let $r_A = \text{rank}(A)$ and $r_B = \text{rank}(B)$, both $m \times n$ matrices. By definition, $r_A = \dim(\text{row space of } A)$, so there is a basis $u_1, \dots, u_{r_A}$ of the row space of A ; similarly let $v_1, \dots, v_{r_B}$ be a basis of the row space of B .

Key observation: every row of $A + B$ lies in the span of $\{u_1, \dots, u_{r_A}, v_1, \dots, v_{r_B}\}$. Indeed, row i of $A + B$ is (row i of A) + (row i of B) – this is exactly how matrix addition works, entry by entry, hence row by row. Row i of A lies in the row space of A , so it is some combination of $u_1, \dots, u_{r_A}$; row i of B lies in the row space of B , so it is some combination of $v_1, \dots, v_{r_B}$. Adding these two combinations gives row i of $A + B$ as a combination of *all* of $u_1, \dots, u_{r_A}, v_1, \dots, v_{r_B}$ together (at most $r_A + r_B$ vectors).

Since this holds for *every* row $i = 1, \dots, m$ of $A + B$, the entire row space of $A + B$ – being spanned by its rows – is contained in the span of these (at most) $r_A + r_B$ vectors $\{u_1, \dots, u_{r_A}, v_1, \dots, v_{r_B}\}$: *why containment of spans follows from containment of generators*: the row space of $A + B$ is by definition the smallest subspace containing all rows of $A + B$; we've shown each such row already lies in $\text{span}\{u_i, v_j\}$, and that span is itself a subspace, so it contains the smallest subspace containing all the rows too.

Bounding the dimension. A subspace spanned by $r_A + r_B$ vectors has dimension at most $r_A + r_B$ (a spanning set's size is always an upper bound for the dimension it spans – if the spanning set were independent, the dimension would equal $r_A + r_B$ exactly; if dependent, some vectors are redundant and the dimension is strictly smaller). Since the row space of $A + B$ is contained in this span, its dimension is at most the dimension of the span it sits inside:

$$\begin{aligned} \text{rank}(A + B) &= \dim(\text{row space of } A + B) \leq \dim(\text{span}\{u_1, \dots, u_{r_A}, v_1, \dots, v_{r_B}\}) \\ &\leq r_A + r_B = \text{rank}(A) + \text{rank}(B). \quad \blacksquare \end{aligned}$$

Why this can be a strict inequality (not always equality). If, say, $B = -A$, then $A + B = 0$, so $\text{rank}(A + B) = 0$, while $\text{rank}(A) + \text{rank}(B) = 2\text{rank}(A)$ can be much larger (whenever $A \neq 0$) – the basis vectors of the two row spaces can overlap or cancel, which is exactly the “redundancy” the spanning-set argument above allows for but does not rule out.

Topic 4 Existence and Uniqueness of Solutions to $Ax = b$

Quick Reference

- $Ax = b$ is **consistent** if it has at least one solution, **inconsistent** otherwise.
- **Theorem 2.6.1** (Existence/Uniqueness): for A an $m \times n$ matrix with $\text{rank}(A) = r$ and $\text{rank}([A \mid b]) = r_a$, exactly one of the following holds: (1) if $r_a = r < n$: infinitely many solutions, of the form $\{u_0 + k_1u_1 + \cdots + k_{n-r}u_{n-r} : k_i \in \mathbb{R}\}$ where $Au_0 = b$ and $Au_i = 0$; (2) if $r_a = r = n$: a unique solution; (3) if $r < r_a$: no solution.
- **Remark 2.6.2**: $Ax = b$ is consistent $\iff \text{rank}(A) = \text{rank}([A \mid b])$.
- **Corollary 2.6.3**: the homogeneous system $Ax = 0$ has a non-trivial solution $\iff \text{rank}(A) < n$.
- If u, v both solve $Ax = b$, then $u - v$ solves the associated homogeneous system $Ax = 0$; so the full solution set of $Ax = b$ (when nonempty) is $\{x_0 + x_h\}$, a particular solution plus every homogeneous solution.

Worked Examples

Example 2.6.1

Suppose $Ax = b$, after Gauss–Jordan reduction, has $[C \mid d] = \begin{pmatrix} 1 & 0 & 2 & -1 & 0 & 0 & 2 & 8 \\ 0 & 1 & 1 & 3 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$

(7 unknowns $x_1, \dots, x_7$). Describe the solution set.

Solution. There are 4 nonzero rows, so $\text{rank}(A) = 4$; the leading terms sit in columns 1, 2, 5, 6, so x_1, x_2, x_5, x_6 are basic and x_3, x_4, x_7 are free – assign them arbitrary constants k_1, k_2, k_3 respectively. Reading off each nonzero row as an equation and solving for the basic variable in terms of the free ones:

$$x_1 + 2x_3 - x_4 + 2x_7 = 8 \Rightarrow x_1 = 8 - 2k_1 + k_2 - 2k_3, \quad x_2 + x_3 + 3x_4 + 5x_7 = 1 \Rightarrow x_2 = 1 - k_1 - 3k_2 - 5k_3,$$

$$x_5 - x_7 = 2 \Rightarrow x_5 = 2 + k_3, \quad x_6 + x_7 = 4 \Rightarrow x_6 = 4 - k_3.$$

Writing the full vector and separating out the part independent of k_1, k_2, k_3 from the parts

multiplying each k_i :

$$x = \begin{pmatrix} 8 \\ 1 \\ 0 \\ 0 \\ 2 \\ 4 \\ 0 \end{pmatrix} + k_1 \begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} 1 \\ -3 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + k_3 \begin{pmatrix} -2 \\ -5 \\ 0 \\ 0 \\ 1 \\ -1 \\ 1 \end{pmatrix} = u_0 + k_1 u_1 + k_2 u_2 + k_3 u_3,$$

matching Theorem 2.6.1's form exactly (with $n - r = 7 - 4 = 3$ free parameters, one for each free variable), where u_0 is a particular solution ($Au_0 = b$, obtained by setting all free variables to 0) and u_1, u_2, u_3 each solve the homogeneous system $Au_i = 0$ (obtained by setting exactly one free variable to 1 and the rest to 0 – since with $b = 0$ on the right, this is exactly what makes each row's equation balance using only that one free variable's contribution).

Practice Questions

Question 4.1

For the system $x + y + z = 3$, $x + 2y + cz = 4$, $2x + 3y + 2cz = k$, find, in terms of the parameters c, k : (i) when there is no solution, (ii) when there is a unique solution, (iii) when there are infinitely many solutions.

Solution. Row-reduce the augmented matrix, keeping c, k symbolic.

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & c & 4 \\ 2 & 3 & 2c & k \end{pmatrix}.$$

Step 1. $R_{21}(-1)$: row 2 $\rightarrow (0, 1, c - 1, 1)$. $R_{31}(-2)$: row 3 $\rightarrow (0, 1, 2c - 2, k - 6)$.

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & c - 1 & 1 \\ 0 & 1 & 2c - 2 & k - 6 \end{pmatrix}.$$

Why we keep $c - 1$ and $2c - 2$ unfactored for now: we don't yet know whether $c - 1 = 0$, so we cannot safely divide by it – we continue with plain row operations (which never require dividing by an unknown) until forced to case-split.

Step 2. $R_{32}(-1)$: row 3 $\rightarrow (0, 0, (2c - 2) - (c - 1), (k - 6) - 1) = (0, 0, c - 1, k - 7)$.

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & c - 1 & 1 \\ 0 & 0 & c - 1 & k - 7 \end{pmatrix}.$$

Why the two occurrences of $c - 1$ are not a coincidence: $2c - 2 = 2(c - 1)$, so subtracting row 2's $(c - 1)$ once from $2(c - 1)$ leaves exactly $(c - 1)$ again – the parameter c enters the third row only through this single combination, which is exactly what will decide the rank.

Case analysis on the pivot candidate $c - 1$ in row 3.

Case A: $c \neq 1$. Then $c - 1 \neq 0$, so it can serve as a third pivot: $\text{rank}(A) = 3 = (\text{number of unknowns})$, and since a pivot exists in every row, $\text{rank}([A \mid b]) = 3$ as well (adding one more

column to a matrix can only ever keep the rank the same or increase it by at most 1; here it cannot increase further since we've already used all 3 rows as pivot rows). By Theorem 2.6.1 case (2) ($r_a = r = n = 3$): **unique solution**, for *every* value of k (the value of k never mattered in this case, since row 3's pivot entry $c - 1$ is nonzero regardless of k).

Case B: $c = 1$. Row 3 becomes $(0, 0, 0, k - 7)$.

- **Case B1:** $k \neq 7$. Row 3 reads $0 = k - 7 \neq 0$ – a contradiction. $\text{rank}(A) = 2$ (only two pivots, in columns 1,2) but $\text{rank}([A \mid b]) = 3$ (the last column, $k - 7 \neq 0$, creates a pivot in the augmented column, since that's the only nonzero entry left in row 3). Since $\text{rank}(A) < \text{rank}([A \mid b])$: **no solution**.
- **Case B2:** $k = 7$. Row 3 is entirely zero: $\text{rank}(A) = \text{rank}([A \mid b]) = 2 < n = 3$. By Theorem 2.6.1 case (1): **infinitely many solutions**, with $n - r = 3 - 2 = 1$ free variable.

Summary.

$$\begin{cases} c \neq 1 : & \text{unique solution, for every } k, \\ c = 1, k \neq 7 : & \text{no solution,} \\ c = 1, k = 7 : & \text{infinitely many solutions.} \end{cases}$$

Sanity check at a concrete point: take $c = 1, k = 7$. The system becomes $x + y + z = 3$, $x + 2y + z = 4$, $2x + 3y + 2z = 7$; indeed the third equation is the sum of the first two ($3 + 4 = 7$, and coefficient-wise $(1, 1, 1) + (1, 2, 1) = (2, 3, 2)$), confirming the redundancy that Case B2 predicts.

Question 4.2

Find the exact condition on a, b, c so that the system $x + 2y - 3z = a$, $2x + 6y - 11z = b$, $x - 2y + 7z = c$ is consistent (Exercise 2.6.6.2).

Solution. Row-reduce $[A \mid b\text{-column}]$ with a, b, c symbolic (careful: here “ b ” is both the name of the second equation's constant *and*, confusingly, the standard name for the right-hand-side vector – we keep them straight by always writing the actual letters a, b, c for the three given constants).

$$\begin{pmatrix} 1 & 2 & -3 & a \\ 2 & 6 & -11 & b \\ 1 & -2 & 7 & c \end{pmatrix}.$$

Step 1. $R_{21}(-2)$: row 2 $\rightarrow (2 - 2, 6 - 4, -11 + 6, b - 2a) = (0, 2, -5, b - 2a)$. $R_{31}(-1)$: row 3 $\rightarrow (0, -4, 10, c - a)$.

$$\begin{pmatrix} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b - 2a \\ 0 & -4 & 10 & c - a \end{pmatrix}.$$

Step 2. To clear the -4 in row 3 using the pivot 2 in row 2: we need $-4 + k(2) = 0 \Rightarrow k = 2$, i.e. $R_{32}(2)$ (add $2 \times$ row 2 to row 3):

row 3 $\rightarrow (0 - 0, -4 + 2(2), 10 + 2(-5), (c - a) + 2(b - 2a)) = (0, 0, 0, c - a + 2b - 4a) = (0, 0, 0, c + 2b - 5a)$.

$$\begin{pmatrix} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b - 2a \\ 0 & 0 & 0 & c + 2b - 5a \end{pmatrix}.$$

Why this row's coefficient part collapses to all zeros: the coefficient matrix $\begin{pmatrix} 1 & 2 & -3 \\ 2 & 6 & -11 \\ 1 & -2 & 7 \end{pmatrix}$ has

rank 2 on its own (the third row's coefficients are already a combination of the first two once you look for it – exactly what this elimination is exposing), so no matter what a, b, c are, the coefficient part of row 3 was always going to vanish; only the constant term $c + 2b - 5a$ survives to possibly break consistency.

Reading off the consistency condition. $\text{rank}(A) = 2$ regardless of a, b, c (two pivots, in columns 1 and 2, are always present since those pivot computations never involved a, b, c). By Remark 2.6.2, the system is consistent exactly when $\text{rank}([A \mid b\text{-col}])$ is *also* 2, i.e. when the third row of the augmented matrix is entirely zero (otherwise the last column creates a third pivot, forcing $\text{rank}([A \mid b\text{-col}]) = 3 > 2 = \text{rank}(A)$, the no-solution case). This means:

$$c + 2b - 5a = 0, \quad \text{i.e.} \quad c = 5a - 2b.$$

Numerical check. Take $a = 1, b = 0$, so the condition demands $c = 5$. The system is $x + 2y - 3z = 1$, $2x + 6y - 11z = 0$, $x - 2y + 7z = 5$; from the reduction above with these values, row 3's constant is $c + 2b - 5a = 5 + 0 - 5 = 0$ ✓, consistent as predicted. Try instead $c = 4 \neq 5$: constant becomes $4 + 0 - 5 = -1 \neq 0$, and indeed row 3 would read $0 = -1$, inconsistent – matching the derived condition.

Question 4.3

Let A be an $n \times n$ matrix. If $A^2x = 0$ has a non-trivial solution, show that $Ax = 0$ also has a non-trivial solution (Exercise 2.6.6.3).

Solution. We argue by contradiction. *Why contradiction is a natural strategy here:* the conclusion we want (“ $Ax = 0$ has a non-trivial solution”) is equivalent, via Corollary 2.6.3, to a statement about rank ($\text{rank}(A) < n$) – and it is often easier to rule out the alternative ($\text{rank}(A) = n$) than to construct a non-trivial solution to $Ax = 0$ directly out of thin air.

Suppose, for contradiction, that $Ax = 0$ has *only* the trivial solution $x = 0$. By Corollary 2.6.3 (applied to A), this means $\text{rank}(A) = n$, i.e. A has full rank. By Question 3.2 (proved above), a square matrix of full rank is row-equivalent to I_n – equivalently (by Theorem 2.7.7, or directly from Theorem 2.6.1 case (2) with $b = 0$), $Ax = 0$ having only the trivial solution is exactly the statement that A is invertible, so A^{-1} exists.

Now let x_0 be *any* solution of $A^2x = 0$, i.e. $A^2x_0 = 0$, which we can also write as $A(Ax_0) = 0$ (matrix multiplication is associative, so $A^2x_0 = A(Ax_0)$ – this is just regrouping, not a new assumption). Since A^{-1} exists, left-multiply both sides of $A(Ax_0) = 0$ by A^{-1} :

$$A^{-1}(A(Ax_0)) = A^{-1}(0) \implies (A^{-1}A)(Ax_0) = 0 \implies I(Ax_0) = 0 \implies Ax_0 = 0.$$

Why each equality above is legitimate: $A^{-1}(0) = 0$ because any matrix times the zero vector is the zero vector; $(A^{-1}A) = I$ is exactly the defining property of the inverse; and $I(Ax_0) = Ax_0$ because the identity matrix does nothing.

So $Ax_0 = 0$ – but we assumed $Ax = 0$ has *only* the trivial solution, forcing $x_0 = 0$. Applying A^{-1} again (or just substituting back): since $x_0 = 0$ was forced for *every* solution x_0 of $A^2x = 0$, the system $A^2x = 0$ can only have the trivial solution too.

This contradicts the given hypothesis that $A^2x = 0$ *has* a non-trivial solution. **The contradiction means our assumption was false:** $Ax = 0$ cannot have only the trivial solution,

i.e. $Ax = 0$ must have a non-trivial solution too. ■

A direct (non-contrapositive) restatement of the same idea, for intuition: if A were invertible, then $A^2 = A \cdot A$ would also be invertible (a product of invertible matrices is invertible), and an invertible matrix's homogeneous system has only the trivial solution – directly contradicting the hypothesis on $A^2x = 0$. So A cannot be invertible, i.e. $\text{rank}(A) < n$, which by Corollary 2.6.3 is exactly the condition for $Ax = 0$ to have a non-trivial solution.

Question 4.4

Solve $x_1 + x_2 + x_3 + x_4 = 4$, $x_1 + 2x_2 + x_3 + 2x_4 = 5$, $2x_1 + 3x_2 + 2x_3 + 3x_4 = 9$, expressing the solution set in the form $u_0 + k_1u_1 + k_2u_2$.

Solution. Augmented matrix:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 4 \\ 1 & 2 & 1 & 2 & 5 \\ 2 & 3 & 2 & 3 & 9 \end{pmatrix}.$$

Step 1. $R_{21}(-1)$: row 2 $\rightarrow$ $(0, 1, 0, 1, 1)$. $R_{31}(-2)$: row 3 $\rightarrow$ $(0, 1, 0, 1, 1)$.

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

Step 2. $R_{32}(-1)$ (rows 2, 3 are now identical, so this produces an all-zero row): row 3 $\rightarrow$ $(0, 0, 0, 0, 0)$.

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Step 3 (back-substitution / finish RREF). $R_{12}(-1)$: row 1 $\rightarrow$ $(1, 0, 1, 0, 3)$.

$$\text{RREF}([A \mid b]) = \begin{pmatrix} 1 & 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Pivot columns are 1, 2 (for x_1, x_2), so x_1, x_2 are basic and x_3, x_4 are free; $\text{rank}(A) = \text{rank}([A \mid b]) = 2$, consistent (Remark 2.6.2), with $n - r = 4 - 2 = 2$ free parameters, matching Theorem 2.6.1's infinite-solution case. Let $x_3 = k_1$, $x_4 = k_2$ (arbitrary). From row 1: $x_1 + x_3 = 3 \Rightarrow x_1 = 3 - k_1$. From row 2: $x_2 + x_4 = 1 \Rightarrow x_2 = 1 - k_2$.

$$x = \begin{pmatrix} 3 - k_1 \\ 1 - k_2 \\ k_1 \\ k_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \end{pmatrix}}_{u_0} + k_1 \underbrace{\begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}}_{u_1} + k_2 \underbrace{\begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}}_{u_2}.$$

Check $Au_0 = b$ (setting $k_1 = k_2 = 0$, i.e. $x = (3, 1, 0, 0)^t$): $3+1+0+0 = 4\checkmark$; $3+2+0+0 = 5\checkmark$; $2(3) + 3(1) + 0 + 0 = 9\checkmark$. **Check** $Au_1 = 0$ (i.e. $x = (-1, 0, 1, 0)^t$, the effect of increasing k_1 by 1 alone): $-1+0+1+0 = 0\checkmark$; $-1+0+1+0 = 0\checkmark$; $2(-1) + 0 + 2(1) + 0 = 0\checkmark$. **Check** $Au_2 = 0$ similarly: $0-1+0+1 = 0\checkmark$; $0-2+0+2 = 0\checkmark$; $0-3+0+3 = 0\checkmark$. All checks pass, confirming both the particular solution and the two homogeneous directions.

Topic 5 Invertible Matrices

Quick Reference

- A (square, order n) is **invertible** if there is B with $AB = BA = I_n$; such B is unique (Lemma 2.7.2) and written A^{-1} .
- **Theorem 2.7.4:** $(A^{-1})^{-1} = A$; $(AB)^{-1} = B^{-1}A^{-1}$ (note the order reverses); $(A^t)^{-1} = (A^{-1})^t$.
- **Theorem 2.7.7** (equivalent conditions): for square A of order n , TFAE: (1) A invertible; (2) $\text{rank}(A) = n$ (“full rank”); (3) A row-equivalent to I_n ; (4) A is a product of elementary matrices.
- **Theorem 2.7.10:** also equivalent to (1)–(4): (5) $Ax = 0$ has only the trivial solution; (6) $Ax = b$ has a solution for every b .
- **Theorem 2.7.8:** a one-sided inverse is already a full inverse – if $AB = I_n$ for some B , or if $CA = I_n$ for some C , then A^{-1} exists (and equals that B , resp. that C).
- **Gauss–Jordan method for A^{-1} :** row reduce $[A \mid I_n]$. If the left block becomes I_n , the right block is A^{-1} ; if the left block ever produces an all-zero row (i.e. never becomes I_n), A is not invertible.

Worked Examples

Example 2.7.13 (Gauss–Jordan inverse)

Find the inverse of $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$.

Solution. Row reduce $[A \mid I_3] = \begin{pmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{pmatrix}$.

$R_1(1/2)$: $\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{pmatrix}$. Then $R_{21}(-1), R_{31}(-1)$: $\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \end{pmatrix}$. Then

$R_2(2/3)$: $\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \end{pmatrix}$. Then $R_{32}(-1/2)$: row 3 $\rightarrow (0, 0, \frac{3}{2} - \frac{1}{6}, -\frac{1}{2} + \frac{1}{6}, -\frac{1}{3}, 1) =$

$(0, 0, \frac{4}{3}, -\frac{1}{3}, -\frac{1}{3}, 1)$. Then $R_3(3/4)$: row 3 $\rightarrow (0, 0, 1, -\frac{1}{4}, -\frac{1}{4}, \frac{3}{4})$.

Back-substitute: $R_{13}(-1/2), R_{23}(-1/3)$: row 1 $\rightarrow (1, \frac{1}{2}, 0, \frac{1}{2} + \frac{1}{8}, \frac{1}{8}, -\frac{3}{8}) = (1, \frac{1}{2}, 0, \frac{5}{8}, \frac{1}{8}, -\frac{3}{8})$; row 2 $\rightarrow (0, 1, 0, -\frac{1}{3} + \frac{1}{12}, \frac{2}{3} + \frac{1}{12}, -\frac{1}{4}) = (0, 1, 0, -\frac{1}{4}, \frac{3}{4}, -\frac{1}{4})$. Finally $R_{12}(-1/2)$: row 1 $\rightarrow (1, 0, 0, \frac{5}{8} + \frac{1}{8}, \frac{1}{8} - \frac{3}{8}, -\frac{3}{8} + \frac{1}{8}) = (1, 0, 0, \frac{3}{4}, -\frac{1}{4}, -\frac{1}{4})$.

$$[I_3 \mid A^{-1}] = \begin{pmatrix} 1 & 0 & 0 & \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}, \quad A^{-1} = \frac{1}{4} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}.$$

Check: $AA^{-1} = \frac{1}{4} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 6-1-1 & -2+3-1 & -2-1+3 \\ 3-2-1 & -1+6-1 & -1-2+3 \\ 3-1-2 & -1+3-2 & -1-1+6 \end{pmatrix} =$

$$\frac{1}{4} \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} = I_3. \checkmark$$

Practice Questions

Question 5.1

Find the inverse of $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \\ 2 & 4 & 7 \end{pmatrix}$ using the Gauss-Jordan method.

Solution. Row reduce $[A \mid I_3] = \begin{pmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 1 & 3 & 2 & 0 & 1 & 0 \\ 2 & 4 & 7 & 0 & 0 & 1 \end{pmatrix}$.

Step 1 (clear column 1 below the pivot, which is already 1). $R_{21}(-1)$: row 2 $\rightarrow$ $(0, 1, -1, -1, 1, 0)$. $R_{31}(-2)$: row 3 $\rightarrow$ $(0, 0, 1, -2, 0, 1)$.

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{pmatrix}.$$

Note: column 1 of row 3 needed no further work after $R_{31}(-2)$ – and, conveniently, the $(3, 2)$ entry is already 0 and the $(3, 3)$ entry is already 1, so row 3 is already fully reduced with pivot in column 3.

Step 2 (back-substitute using row 3 to clear column 3 above it). $R_{13}(-3)$: row 1 $\rightarrow$ $(1, 2, 0, 1 - 3(-2), 0 - 3(0), 0 - 3(1)) = (1, 2, 0, 7, 0, -3)$. $R_{23}(1)$: row 2 $\rightarrow$ $(0, 1, 0, -1 + (-2), 1 + 0, 0 + 1) = (0, 1, 0, -3, 1, 1)$.

$$\begin{pmatrix} 1 & 2 & 0 & 7 & 0 & -3 \\ 0 & 1 & 0 & -3 & 1 & 1 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{pmatrix}.$$

Step 3 (clear column 2 above its pivot). $R_{12}(-2)$: row 1 $\rightarrow$ $(1, 0, 0, 7 - 2(-3), 0 - 2(1), -3 - 2(1)) = (1, 0, 0, 13, -2, -5)$.

$$[I_3 \mid A^{-1}] = \begin{pmatrix} 1 & 0 & 0 & 13 & -2 & -5 \\ 0 & 1 & 0 & -3 & 1 & 1 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} 13 & -2 & -5 \\ -3 & 1 & 1 \\ -2 & 0 & 1 \end{pmatrix}.$$

Why we are entitled to read off A^{-1} directly from the right-hand block: the whole point of Gauss-Jordan reduction on $[A \mid I]$ is that every row operation applied is simultaneously a left-multiplication by some elementary matrix (Question 2.2's fact), and the *same sequence* of operations is applied to both halves at once. If that sequence takes A all the way to I , i.e. $E_k \cdots E_1 A = I$, then the identical operations applied to the right half give $E_k \cdots E_1 I = E_k \cdots E_1$ – but $E_k \cdots E_1$ is exactly the matrix that turns A into I , i.e. $E_k \cdots E_1 = A^{-1}$ by

definition of the inverse. So the right-hand block, once the left-hand block reaches I , is nothing but A^{-1} itself (Corollary 2.7.12).

Check: $AA^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \\ 2 & 4 & 7 \end{pmatrix} \begin{pmatrix} 13 & -2 & -5 \\ -3 & 1 & 1 \\ -2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 13-6-6 & -2+2+0 & -5+2+3 \\ 13-9-4 & -2+3+0 & -5+3+2 \\ 26-12-14 & -4+4+0 & -10+4+7 \end{pmatrix} =$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3. \checkmark$$

Question 5.2

If A is a symmetric invertible matrix (i.e. $A^t = A$), is A^{-1} also symmetric?

Solution. We must check whether $(A^{-1})^t = A^{-1}$.

By Theorem 2.7.4, part 3, for *any* invertible matrix A : $(A^t)^{-1} = (A^{-1})^t$. *Why this identity holds in general* (a quick reminder, since we use it as a black box here): taking the transpose of $AA^{-1} = I$ and $A^{-1}A = I$ gives $(A^{-1})^t A^t = I^t = I$ and $A^t (A^{-1})^t = I$ – i.e. $(A^{-1})^t$ is a two-sided inverse of A^t , and by uniqueness of inverses (Lemma 2.7.2) this forces $(A^t)^{-1} = (A^{-1})^t$.

Now use the hypothesis $A^t = A$ (symmetry of A) to substitute directly into the left side of this identity:

$$(A^{-1})^t = (A^t)^{-1} = (A)^{-1} = A^{-1}.$$

Where each step comes from: the first equality is Theorem 2.7.4.3 (true for every invertible matrix); the second substitutes $A^t = A$ (this is exactly, and only, where the symmetry hypothesis is used); the last is just notation.

Since $(A^{-1})^t = A^{-1}$, **yes, A^{-1} is symmetric.** ■

Numerical illustration. Take the symmetric matrix from Example 2.7.13, $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$

($A^t = A$ by inspection, since it is symmetric about the diagonal). Its inverse was computed

there as $A^{-1} = \frac{1}{4} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}$, which is indeed symmetric about its own diagonal too – exactly as the proof predicts.

Question 5.3

Show that every elementary matrix is invertible, and that its inverse is again an elementary matrix of the *same type* (Exercise 2.7.5.2).

Solution. There are three types to check, one for each kind of elementary row operation; in each case we exhibit an explicit two-sided inverse and note it has the same type.

Type E_{ij} (row swap). E_{ij} is obtained from I_n by swapping rows i, j . Swapping rows i, j of I_n *twice* returns I_n exactly to itself (the second swap exactly undoes the first, since swapping the same two rows twice is the identity operation on rows). Algebraically, this says $E_{ij}E_{ij} = I_n$ (left-multiplying E_{ij} by E_{ij} again performs the same row-swap operation on E_{ij} as a matrix, which by the same logic returns I_n). So E_{ij} is its own two-sided inverse: $E_{ij}^{-1} = E_{ij}$, itself an elementary matrix of type E_{ij} (the same type, with the same indices).

Type $E_k(c)$ (row scaling, $c \neq 0$). $E_k(c)$ scales row k of I_n by c . Consider $E_k(1/c)$ (well-

defined precisely because $c \neq 0$, so $1/c$ exists), which scales row k by $1/c$. Applying $E_k(1/c)$ to $E_k(c)$ (i.e. computing $E_k(1/c)E_k(c)$) scales row k – which is already scaled by c from I_n 's original row k – by a further $1/c$, giving $c \cdot \frac{1}{c} = 1$ times the original row k of I_n , i.e. row k of I_n unchanged; all other rows were never touched by either operation. So $E_k(1/c)E_k(c) = I_n$, and by the same reasoning in the other order $E_k(c)E_k(1/c) = I_n$. Thus $E_k(c)^{-1} = E_k(1/c)$ – again an elementary matrix of the *same type* $E_k(\cdot)$ (just with parameter $1/c$ instead of c ; note $1/c \neq 0$ since $c \neq 0$, so this is a legitimate elementary matrix of this type).

Type $E_{ij}(c)$ (add $c \times \text{row } j$ to row i , $i \neq j$). Consider $E_{ij}(-c)$, which adds $-c$ times row j to row i . Apply $E_{ij}(-c)$ after $E_{ij}(c)$: starting from I_n , first row i becomes (row i) $+ c \cdot$ (row j); since $i \neq j$, row j itself is untouched by this first operation. Then applying $E_{ij}(-c)$ to the result adds $-c \times$ (row j , still unchanged) to the *new* row i : $[(\text{row } i) + c(\text{row } j)] + (-c)(\text{row } j) = \text{row } i$, exactly cancelling the earlier addition and restoring row i to its original value. So $E_{ij}(-c)E_{ij}(c) = I_n$ (and symmetrically the other order), giving $E_{ij}(c)^{-1} = E_{ij}(-c)$ – again elementary, of the same type $E_{ij}(\cdot)$, with parameter $-c$ instead of c .

Conclusion. In every case, an elementary matrix has an explicit two-sided inverse that is again elementary, of the same type (same row-index data, with the scaling parameter inverted or negated as appropriate). So every elementary matrix is invertible, **and its inverse is always another elementary matrix** – indeed, of the very same type it started as. ■

Question 5.4

Show that a triangular matrix A (order n) is invertible if and only if every diagonal entry of A is nonzero (Exercise 2.7.11.1).

Solution. We treat the upper triangular case in detail ($a_{ij} = 0$ whenever $i > j$); the lower triangular case then follows by the transpose trick noted at the end.

($\Leftarrow$) **Suppose every diagonal entry $a_{11}, \dots, a_{nn}$ is nonzero.** We show A can be row-reduced to I_n using only the diagonal entries as pivots, without ever needing to swap rows. Proceed column by column, $k = 1, 2, \dots, n$ in order: when we reach column k , all entries *below* the diagonal in columns $1, \dots, k-1$ have already been cleared to 0 by the previous steps, and since A is upper triangular, all entries below the diagonal in column k itself, i.e. a_{ik} for $i > k$, are *already* 0 to begin with (that is exactly what upper triangular means) – so there is nothing to eliminate below row k in column k ; only the diagonal entry $a_{kk} \neq 0$ needs to be normalized to 1 via $R_k(1/a_{kk})$ (legitimate exactly because $a_{kk} \neq 0$). After all n columns are processed this way, we have a matrix with 1's on the diagonal and 0's below – upper triangular with unit diagonal – and a further back-substitution pass (clearing entries above each pivot, working from the last pivot backward) reduces it fully to I_n , using only the already-established 1's as pivots (no new zero-pivot issue can arise, since the diagonal is already all 1's at this stage).

Since A is row-equivalent to I_n this way, and every step used was a legitimate elementary row operation with a nonzero pivot at each stage (no division by zero, no need for a swap), Theorem 2.7.7 (specifically, condition (3) $\Rightarrow$ (1)) gives that A is invertible.

($\Rightarrow$) **Suppose instead some diagonal entry is zero**, say $a_{kk} = 0$ for some k ; we show A is *not* invertible, by exhibiting $\text{rank}(A) < n$.

Consider rows $k, k+1, \dots, n$ of A – there are $n-k+1$ such rows. **Claim: each of these rows has a 0 in every one of columns $1, \dots, k$.** For a row i with $i > k$: since A is upper triangular and $i > k$, in particular $i > j$ for every $j \leq k$, so $a_{ij} = 0$ for all $j = 1, \dots, k$ – this is immediate from the upper-triangular condition. For row $i = k$ itself: a_{kj} for $j < k$ is 0 because... actually

upper triangular only guarantees zeros *below* the diagonal ($i > j$), and here $i = k > j$ is exactly that case for $j < k$, so $a_{kj} = 0$ for $j < k$; and $a_{kk} = 0$ by our assumption. So indeed row k also has zeros in all of columns $1, \dots, k$.

So all $n - k + 1$ of these rows lie entirely within the subspace $\{x \in \mathbb{R}^n : x_1 = x_2 = \dots = x_k = 0\}$, which has dimension $n - k$ (it is spanned by the $n - k$ standard basis vectors $e_{k+1}, \dots, e_n$). But we have $n - k + 1$ vectors – one more than the dimension of the space they live in. *Why more vectors than the dimension forces dependence*: this is exactly the basic fact (used repeatedly throughout this course) that any set of vectors in a space, with more vectors than the space's dimension, must be linearly dependent (otherwise they would be an independent spanning-adjacent set exceeding a basis's size, which extending to a basis would make too large). So rows $k, \dots, n$ of A are linearly dependent.

A dependency among a subset of A 's rows immediately gives a dependency among *all* of A 's rows (pad the dependency with zero coefficients on the remaining rows $1, \dots, k - 1$). So the full set of n rows of A is linearly dependent, meaning $\text{rank}(A) < n$ (a full set of n independent rows would be required for rank n ; dependence means the row space has dimension strictly less than n). By Theorem 2.7.7 (condition (2) fails, since A is not of full rank), A is **not** invertible.

Lower triangular case. If A is lower triangular, then A^t is upper triangular, with the *same* diagonal entries as A (transposing doesn't move diagonal entries). By the case just proved, A^t is invertible iff every diagonal entry of A^t (equivalently, of A) is nonzero. And A is invertible iff A^t is invertible (since $(A^t)^{-1} = (A^{-1})^t$ exists exactly when A^{-1} does, by Theorem 2.7.4.3) – so A is invertible iff every diagonal entry of A is nonzero, exactly as claimed. ■

Question 5.5

If P, Q are invertible matrices (of appropriate sizes) and PAQ is defined, show that $\text{rank}(PAQ) = \text{rank}(A)$ (Exercise 2.7.5.4).

Solution. We show $\text{rank}(PA) = \text{rank}(A)$ first (left multiplication alone), then $\text{rank}(AQ) = \text{rank}(A)$ (right multiplication alone), and combine.

Left multiplication by invertible P preserves rank. By Theorem 2.7.7 (condition (1) $\Rightarrow$ (4)), an invertible matrix P is a product of elementary matrices, $P = E_1 E_2 \dots E_k$. So $PA = E_1 E_2 \dots E_k A$, which (reading right to left, one elementary matrix at a time, exactly as in Question 2.3) is the result of applying a *sequence of elementary row operations* to A . But elementary row operations are, by construction, exactly the operations that define row-equivalence – and row-equivalent matrices always have the same rank (row-rank is invariant under row operations, since the RREF reached is the same regardless of the path taken, by uniqueness of RREF, Theorem 2.4.11). So $\text{rank}(PA) = \text{rank}(A)$.

Right multiplication by invertible Q preserves rank. By Question 3.1, $\text{rank}(M) = \text{rank}(M^t)$ for any matrix M . Apply this together with the left-multiplication result just proved: $\text{rank}(AQ) = \text{rank}((AQ)^t) = \text{rank}(Q^t A^t)$. Now Q^t is invertible whenever Q is (since $(Q^t)^{-1} = (Q^{-1})^t$ exists), so by the left-multiplication case applied to Q^t and A^t : $\text{rank}(Q^t A^t) = \text{rank}(A^t) = \text{rank}(A)$ (the last equality again by Question 3.1). Chaining these equalities: $\text{rank}(AQ) = \text{rank}(A)$.

Combining both. $\text{rank}(PAQ) = \text{rank}(P(AQ))$; treating AQ as a single matrix and applying the left-multiplication result to it: $\text{rank}(P(AQ)) = \text{rank}(AQ)$. And we just showed $\text{rank}(AQ) = \text{rank}(A)$. Chaining:

$$\text{rank}(PAQ) = \text{rank}(AQ) = \text{rank}(A). \quad \blacksquare$$

Why invertibility of P, Q is essential (not just a convenient hypothesis): if P or Q were allowed to be singular, multiplying by them can only ever *decrease* rank, never increase it, and can decrease it strictly – e.g. $P = 0$ gives $PA = 0$, rank 0, regardless of $\text{rank}(A)$. It is exactly the reversibility of an invertible matrix's corresponding row/column operations (each one has an inverse operation, Question 5.3) that guarantees no information – and hence no rank – is ever lost or gained.

Topic 6 Determinants, Adjoint, and Cramer's Rule

Quick Reference

- For a square matrix A of order n , $A(i|j)$ denotes the submatrix with row i and column j deleted. The (i, j) -**minor** is $A_{ij} = \det(A(i|j))$; the (i, j) -**cofactor** is $C_{ij} = (-1)^{i+j} A_{ij}$.
- **Determinant** (Definition 2.8.2, expansion along row 1): $\det([a]) = a$ for 1×1 ; otherwise $\det(A) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A(1|j))$. **Remark 2.8.8:** expansion along *any* row k gives the same value: $\det(A) = \sum_j (-1)^{k+j} a_{kj} \det(A(k|j))$.
- Key properties (Lemma 2.8.7, used freely below): swapping two rows negates $\det$; a repeated row forces $\det = 0$; $\det(AB) = \det(A) \det(B)$ (Theorem 2.8.15); $\det(A) = \det(A^t)$ (Theorem 2.8.17); $\det(A) \neq 0 \iff A$ invertible (Corollary 2.8.16, essentially Theorem 2.7.7 with determinants added to the list).
- **Adjoint:** $\text{Adj}(A) = [b_{ij}]$ with $b_{ij} = C_{ji}$ (the transpose of the cofactor matrix). **Theorem 2.8.12:** $A \cdot \text{Adj}(A) = \det(A) I_n$, so if $\det(A) \neq 0$, $A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$.
- **Cramer's Rule** (Theorem 2.8.18): for $Ax = b$ with $\det(A) \neq 0$, the unique solution is $x_j = \det(A_j) / \det(A)$, where A_j is A with column j replaced by b .

Worked Examples

Example 2.8.4 (cofactor expansion, 2×2)

For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, find $\det(A)$.

Solution. By definition (expansion along row 1, the only option available for the base 2×2 formula): $\det(A) = a_{11}a_{22} - a_{12}a_{21} = 1(1) - 2(2) = 1 - 4 = -3$.

Example 2.8.11 (Adjoint)

For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$, find $\text{Adj}(A)$.

Solution. Compute all nine cofactors $C_{ij} = (-1)^{i+j} A_{ij}$ (deleting row i , column j , then taking

the 2×2 determinant, then applying the sign):

$$C_{11} = +\det \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} = 6 - 2 = 4, \quad C_{12} = -\det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = -(4 - 1) = -3,$$

$$C_{13} = +\det \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} = 4 - 3 = 1, \quad C_{21} = -\det \begin{pmatrix} 2 & 3 \\ 2 & 2 \end{pmatrix} = -(4 - 6) = 2,$$

$$C_{22} = +\det \begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix} = 2 - 3 = -1, \quad C_{23} = -\det \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} = -(2 - 2) = 0,$$

$$C_{31} = +\det \begin{pmatrix} 2 & 3 \\ 3 & 1 \end{pmatrix} = 2 - 9 = -7, \quad C_{32} = -\det \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix} = -(1 - 6) = 5,$$

$$C_{33} = +\det \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = 3 - 4 = -1.$$

$\text{Adj}(A)$ has (i, j) entry C_{ji} – i.e. it is the *transpose* of the cofactor matrix $[C_{ij}]$:

$$[C_{ij}] = \begin{pmatrix} 4 & -3 & 1 \\ 2 & -1 & 0 \\ -7 & 5 & -1 \end{pmatrix} \implies \text{Adj}(A) = [C_{ij}]^t = \begin{pmatrix} 4 & 2 & -7 \\ -3 & -1 & 5 \\ 1 & 0 & -1 \end{pmatrix}.$$

Example 2.8.13 (inverse via adjoint)

For $A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}$, find A^{-1} using the adjoint.

Solution. First, $\det(A)$ by expansion along row 1: $\det(A) = 1 \cdot \det \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} - (-1) \cdot \det \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} + 0 = 1(1 - 2) + 1(0 - 1) + 0 = -1 - 1 = -2$.

Since $\det(A) = -2 \neq 0$, A is invertible. Computing the cofactors (as in the previous example)

gives $\text{Adj}(A) = \begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -3 & 1 \end{pmatrix}$, so by Theorem 2.8.12.3,

$$A^{-1} = \frac{1}{\det(A)} \text{Adj}(A) = \frac{1}{-2} \begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -3 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} \end{pmatrix}.$$

Example 2.8.19 (Cramer's Rule)

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$ and $b = (1, 1, 1)^t$. Find x with $Ax = b$ using Cramer's rule.

Solution. First $\det(A)$ (this is the same A as in Example 2.8.11, so we may reuse those cofactors: expanding along row 1, $\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = 1(4) + 2(-3) + 3(1) = 4 - 6 + 3 = 1$). Since $\det(A) = 1 \neq 0$, Cramer's rule applies. Form A_1, A_2, A_3 by replacing the

respective column of A with b :

$$\det(A_1) = \det \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} = 1(6 - 2) - 2(2 - 1) + 3(2 - 3) = 4 - 2 - 3 = -1,$$

$$\det(A_2) = \det \begin{pmatrix} 1 & 1 & 3 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} = 1(2 - 1) - 1(4 - 1) + 3(2 - 1) = 1 - 3 + 3 = 1,$$

$$\det(A_3) = \det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 1 \end{pmatrix} = 1(3 - 2) - 2(2 - 1) + 1(4 - 3) = 1 - 2 + 1 = 0.$$

So $x_1 = \det(A_1)/\det(A) = -1/1 = -1$, $x_2 = 1/1 = 1$, $x_3 = 0/1 = 0$: $x^t = (-1, 1, 0)$.

Check: $Ax = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 + 2 + 0 \\ -2 + 3 + 0 \\ -1 + 2 + 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = b. \checkmark$

Practice Questions

Question 6.1

Compute $\det(M)$ for $M = \begin{pmatrix} 2 & 0 & 3 \\ 1 & 4 & -1 \\ 3 & 2 & 5 \end{pmatrix}$, choosing the expansion row wisely.

Solution. We may expand along *any* row or column and get the same answer (Remark 2.8.8) – so we choose row 1, since it contains a 0 in position $(1, 2)$, meaning we can skip computing that minor entirely (its coefficient in the expansion is 0, so whatever the minor is, it contributes nothing).

By definition, expanding along row 1:

$$\det(M) = (-1)^{1+1}m_{11}\det(M(1|1)) + (-1)^{1+2}m_{12}\det(M(1|2)) + (-1)^{1+3}m_{13}\det(M(1|3)).$$

The middle term has $m_{12} = 0$, so it vanishes regardless of the minor – we don't even need to write out $M(1|2)$. Computing the two surviving minors:

$$M(1|1) = \begin{pmatrix} 4 & -1 \\ 2 & 5 \end{pmatrix} \text{ (delete row 1, column 1),} \quad \det(M(1|1)) = 4(5) - (-1)(2) = 20 + 2 = 22,$$

$$M(1|3) = \begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix} \text{ (delete row 1, column 3),} \quad \det(M(1|3)) = 1(2) - 4(3) = 2 - 12 = -10.$$

So:

$$\det(M) = (+1)(2)(22) + 0 + (+1)(3)(-10) = 44 - 30 = 14.$$

Why the sign on the third term is +, not -: the cofactor sign is $(-1)^{i+j} = (-1)^{1+3} = (-1)^4 = +1$ (an even power), whereas the middle term's sign would have been $(-1)^{1+2} = -1$ (odd) – but since its coefficient $m_{12} = 0$ anyway, the sign there was irrelevant.

Cross-check by expanding along column 2 instead, which also has one zero entry ($m_{12} = 0$) plus generally different-looking minors, as a genuine independent check that the value 14 is not an arithmetic accident of one particular expansion:

$$\det(M) = (-1)^{1+2}m_{12}\det(M(1|2)) + (-1)^{2+2}m_{22}\det(M(2|2)) + (-1)^{3+2}m_{32}\det(M(3|2)).$$

First term vanishes ($m_{12} = 0$). Second: $M(2|2) = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$, $\det = 10 - 9 = 1$, sign $(-1)^4 = +1$, contributes $4(1) = 4$. Third: $M(3|2) = \begin{pmatrix} 2 & 3 \\ 1 & -1 \end{pmatrix}$, $\det = -2 - 3 = -5$, sign $(-1)^5 = -1$, contributes $-1 \cdot 2 \cdot (-5) = 10$. Total: $0 + 4 + 10 = 14$. ✓ Matches exactly, as Remark 2.8.8 guarantees it must.

Question 6.2

Prove that the determinant of an $n \times n$ upper triangular matrix equals the product of its diagonal entries.

Solution. We argue by induction on n , the size of the matrix. *Why induction is the natural tool:* the definition of the determinant is itself inductive (an $n \times n$ determinant is defined in terms of $(n-1) \times (n-1)$ determinants), so a proof by induction on n can directly mirror that recursive structure.

Base case $n = 1$. $A = [a_{11}]$, and by definition $\det(A) = a_{11}$, which is trivially “the product of its diagonal entries” (there being only one).

Inductive hypothesis. Assume the claim holds for all upper triangular matrices of size $(n-1) \times (n-1)$: their determinant is the product of their diagonal entries.

Inductive step. Let A be $n \times n$ upper triangular, so $a_{ij} = 0$ whenever $i > j$; in particular, column 1 of A has a_{11} possibly nonzero in row 1, but $a_{i1} = 0$ for every $i > 1$ (since $i > 1 = j$ satisfies $i > j$). Expand $\det(A)$ along *column 1* (a valid choice, since expansion along any row *or* column gives the determinant – this extension of Remark 2.8.8 to columns follows from $\det(A) = \det(A^t)$, expanding A^t along its row 1 being the same as expanding A along its column 1):

$$\det(A) = \sum_{i=1}^n (-1)^{i+1} a_{i1} \det(A(i|1)).$$

Every term with $i > 1$ has $a_{i1} = 0$ (shown above), so it vanishes regardless of the minor. Only the $i = 1$ term survives:

$$\det(A) = (-1)^{1+1} a_{11} \det(A(1|1)) = a_{11} \det(A(1|1)).$$

Now, $A(1|1)$ is A with row 1 and column 1 deleted – i.e. the $(n-1) \times (n-1)$ submatrix formed by rows $2, \dots, n$ and columns $2, \dots, n$ of A . *Why this submatrix is again upper triangular, with the remaining diagonal entries of A as its own diagonal:* an entry of $A(1|1)$ in its own position (i', j') (for $1 \leq i', j' \leq n-1$) corresponds to entry $(i'+1, j'+1)$ of the original A ; if $i' > j'$ then $i'+1 > j'+1$ too, so this entry is 0 by A 's upper-triangularity – meaning $A(1|1)$ is itself upper triangular. And its diagonal entries, at (i', i') for $i' = 1, \dots, n-1$, correspond to A 's original entries at $(i'+1, i'+1)$, i.e. exactly $a_{22}, a_{33}, \dots, a_{nn}$ – all of A 's diagonal entries *except* a_{11} .

By the inductive hypothesis, applied to this $(n-1) \times (n-1)$ upper triangular matrix $A(1|1)$:

$$\det(A(1|1)) = a_{22}a_{33} \cdots a_{nn}.$$

Substituting back:

$$\det(A) = a_{11} \cdot \det(A(1|1)) = a_{11} \cdot (a_{22}a_{33} \cdots a_{nn}) = a_{11}a_{22} \cdots a_{nn},$$

which is exactly the product of all n diagonal entries of A . By induction, the claim holds for every $n \geq 1$. ■

Remark (lower triangular case). The identical statement holds for lower triangular matrices too, either by an entirely symmetric induction (expanding along row 1 instead of column 1, since a lower triangular matrix's row 1 has only a_{11} possibly nonzero), or immediately from the upper-triangular case via $\det(A) = \det(A^t)$ (a lower triangular A has upper triangular A^t , with the same diagonal entries).

Question 6.3

Let A be an orthogonal matrix, i.e. $A^t A = I$. Show that $\det(A) = \pm 1$ (Exercise 2.9.1.1).

Solution. Take determinants of both sides of $A^t A = I$. *Why we're allowed to just "take determinants of both sides":* if two matrices are equal, they are literally the same matrix, so any function evaluated at one gives the same output evaluated at the other – in particular $\det(A^t A) = \det(I)$.

Left side. By Theorem 2.8.15, $\det(A^t A) = \det(A^t) \det(A)$. By Theorem 2.8.17, $\det(A^t) = \det(A)$. Substituting: $\det(A^t A) = \det(A) \cdot \det(A) = \det(A)^2$.

Right side. $\det(I) = 1$ (the identity matrix is upper – and lower – triangular with every diagonal entry 1, so by Question 6.2, $\det(I) = 1 \cdot 1 \cdots 1 = 1$; or simply by direct computation for any size).

Equating: $\det(A)^2 = 1$. *Why this forces $\det(A) = \pm 1$ and nothing else:* $\det(A)$ is a single real number (determinants of real matrices are real numbers), and the only real numbers whose square is 1 are 1 and -1 (if $t^2 = 1$ then $t^2 - 1 = 0$, i.e. $(t - 1)(t + 1) = 0$, so $t = 1$ or $t = -1$ – a product of real numbers is zero only if one of the factors is zero). Hence $\det(A) = \pm 1$. ■

Remark. Both signs genuinely occur: I_n itself is orthogonal with $\det(I_n) = 1$, while swapping two rows of I_n (giving an elementary matrix E_{ij} , which satisfies $E_{ij}^t E_{ij} = E_{ij} E_{ij} = I$ by Question 5.3, so E_{ij} is orthogonal too) has $\det(E_{ij}) = -1$ (a single row swap negates the determinant of I , by the row-swap property in the Quick Reference).

Question 6.4

For an $n \times n$ matrix A , show that the following are all equivalent: (a) A is singular (A^{-1} does not exist); (b) $\text{rank}(A) < n$; (c) $\det(A) = 0$; (d) A is not row-equivalent to I_n ; (e) $Ax = 0$ has a non-trivial solution; (f) $Ax = b$ does not have a unique solution for some b (Exercise 2.9.1.3, with statement (b) corrected to the mathematically consistent form – “singular” must correspond to *less than* full rank, not equal to n).

Solution. We already have almost all of this machinery assembled from earlier topics; the only genuinely new ingredient is linking $\det(A) = 0$ into the chain, via Theorem 2.8.12 (adjoint) and Theorem 2.8.15 (multiplicativity of determinant), which is the one fact from this Topic not yet used elsewhere. We prove a cycle of implications $(a) \Rightarrow (c) \Rightarrow (b) \Rightarrow (d) \Rightarrow (e) \Rightarrow (f) \Rightarrow (a)$, which is enough to establish that all six are equivalent (each one implies the next, and the last loops back to the first, so any two of them imply each other by following the chain around).

(a) $\Rightarrow$ (c): singular $\Rightarrow \det(A) = 0$. Suppose, for contradiction, $\det(A) \neq 0$. Then Theorem

2.8.12.3 gives $A \cdot (\frac{1}{\det(A)} \text{Adj}(A)) = I$, and one checks symmetrically $(\frac{1}{\det(A)} \text{Adj}(A)) \cdot A = I$ as well (Corollary 2.8.14 handles this other order) – so $\frac{1}{\det(A)} \text{Adj}(A)$ is a genuine two-sided inverse of A , contradicting that A is singular. So $\det(A) = 0$.

(c)⇒(b): $\det(A) = 0 \Rightarrow \text{rank}(A) < n$. Contrapositive: suppose $\text{rank}(A) = n$. By Question 3.2, A is row-equivalent to I_n , say via elementary matrices $E_1, \dots, E_k$ with $E_k \cdots E_1 A = I_n$. Taking determinants and using $\det(E_k \cdots E_1 A) = \det(E_k) \cdots \det(E_1) \det(A)$ (repeated application of Theorem 2.8.15): $\det(E_k) \cdots \det(E_1) \det(A) = \det(I_n) = 1$. Each $\det(E_i) \neq 0$ (a row-swap has $\det = -1 \neq 0$; a row-scaling $E_k(c)$ by $c \neq 0$ has determinant $c \neq 0$; an $E_{ij}(c)$ is triangular with all diagonal entries 1, so by Question 6.2 its determinant is $1 \neq 0$) – so the product $\det(E_k) \cdots \det(E_1)$ is a nonzero real number, and since the whole product with $\det(A)$ equals $1 \neq 0$, we must have $\det(A) \neq 0$ too (if $\det(A)$ were 0, the whole left side would be $0 \neq 1$). Contrapositive established: $\det(A) = 0 \Rightarrow \text{rank}(A) < n$.

(b)⇒(d): $\text{rank}(A) < n \Rightarrow A$ **not row-equivalent to I_n** . If A were row-equivalent to I_n , then (row-equivalent matrices share the same rank, since row operations don't change row-rank) $\text{rank}(A) = \text{rank}(I_n) = n$, contradicting $\text{rank}(A) < n$.

(d)⇒(e): not row-equivalent to $I_n \Rightarrow Ax = 0$ has a non-trivial solution. Contrapositive of Question 3.2 combined with the ($\Leftarrow$) direction implicit in the full Theorem 2.7.7 chain: if $Ax = 0$ had *only* the trivial solution, then (by Corollary 2.6.3's contrapositive) $\text{rank}(A) = n$, and then by Question 3.2, A would be row-equivalent to I_n – contradicting (d). So $Ax = 0$ must have a non-trivial solution.

(e)⇒(f): $Ax = 0$ has a non-trivial solution $\Rightarrow Ax = b$ fails to have a unique solution for some b . Take $b = 0$ itself: $Ax = 0$ has *at least two* solutions (the trivial one $x = 0$, and the given non-trivial one), so it certainly does not have a *unique* solution. This $b = 0$ is the required example.

(f)⇒(a): some b gives a non-unique (or no) solution $\Rightarrow A$ is singular. Contrapositive: suppose A is invertible. Then for *every* b , $x = A^{-1}b$ solves $Ax = b$ (since $A(A^{-1}b) = (AA^{-1})b = Ib = b$), and it is the *only* solution (if $Ax' = b$ too, left-multiply by A^{-1} : $x' = A^{-1}b = x$). So an invertible A gives a unique solution for *every* b – contradicting the hypothesis that *some* b fails to have a unique solution. So A must be singular.

Closing the cycle. We've shown $(a) \Rightarrow (c) \Rightarrow (b) \Rightarrow (d) \Rightarrow (e) \Rightarrow (f) \Rightarrow (a)$. Since this is a full cycle, starting anywhere and following the arrows reaches every other statement – e.g. $(b) \Rightarrow (d) \Rightarrow (e) \Rightarrow (f) \Rightarrow (a) \Rightarrow (c) \Rightarrow (b)$ shows (b) implies, and is implied by, every other statement too. Hence all six statements (a)–(f) are equivalent. ■

Why this particular cycle order was chosen (rather than, say, proving all $\binom{6}{2} = 15$ pairwise implications directly): a single cycle through all n statements requires only n implications to establish full equivalence, each one reusing a specific tool already built in an earlier topic (rank, row-equivalence, determinants, homogeneous systems, uniqueness) – this is both the shortest route and the one that best shows how every tool from this chapter fits into one unified picture.

Question 6.5

(Vandermonde determinant, 3×3 case of Exercise 2.9.1.5.) Show that for any a, b, c ,

$$\det \begin{pmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{pmatrix} = (b-a)(c-a)(c-b).$$

Solution. Apply elementary *column* operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$ (subtracting column 1 from columns 2 and 3). *Why this is a legitimate way to compute the determinant:* column operations of this type (replace a column by itself plus a multiple of another column) leave the determinant *unchanged* – this is the column-analogue of the corresponding row property in the Quick Reference (itself following from $\det(A) = \det(A^t)$: a column operation on A is a row operation on A^t , which by the row version doesn't change $\det(A^t) = \det(A)$). Column 2 becomes $(1-1, b-a, b^2-a^2)^t = (0, b-a, b^2-a^2)^t$, and b^2-a^2 factors as $(b-a)(b+a)$ (difference of squares) – so column 2 is $(0, b-a, (b-a)(b+a))^t$. Similarly column 3 becomes $(0, c-a, (c-a)(c+a))^t$. The determinant, unchanged by these operations, is now

$$\det \begin{pmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & (b-a)(b+a) & (c-a)(c+a) \end{pmatrix}.$$

Expand along row 1 (it now has two zeros, in positions (1, 2) and (1, 3), so only the (1, 1) term survives – exactly the same time-saving idea as Question 6.1):

$$= 1 \cdot \det \begin{pmatrix} b-a & c-a \\ (b-a)(b+a) & (c-a)(c+a) \end{pmatrix}.$$

Factor $(b-a)$ out of column 1 and $(c-a)$ out of column 2 of this 2×2 determinant. *Why a common factor can be pulled out of an entire column like this:* a determinant is linear in each column separately (this is part of the multilinearity underlying the whole cofactor-expansion definition) – concretely here, both entries of column 1 share the factor $(b-a)$ (the top entry *is* $b-a$, and the bottom entry is $(b-a)(b+a)$), so column 1 = $(b-a) \cdot (1, b+a)^t$, and pulling a common scalar factor out of a column scales the whole determinant by that factor. Doing this for both columns:

$$= (b-a)(c-a) \det \begin{pmatrix} 1 & 1 \\ b+a & c+a \end{pmatrix} = (b-a)(c-a)[(c+a) - (b+a)] = (b-a)(c-a)(c-b).$$

This is exactly the claimed formula. ■

Numerical check: take $a = 1, b = 2, c = 3$. The formula gives $(2-1)(3-1)(3-2) = 1 \cdot 2 \cdot 1 = 2$.

Direct computation of $\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$ (expanding along row 1): $1(2 \cdot 9 - 3 \cdot 4) - 1(1 \cdot 9 - 3 \cdot 1) + 1(1 \cdot 4 - 2 \cdot 1) = 1(18 - 12) - 1(9 - 3) + 1(4 - 2) = 6 - 6 + 2 = 2. \checkmark$